



Autumn 2025

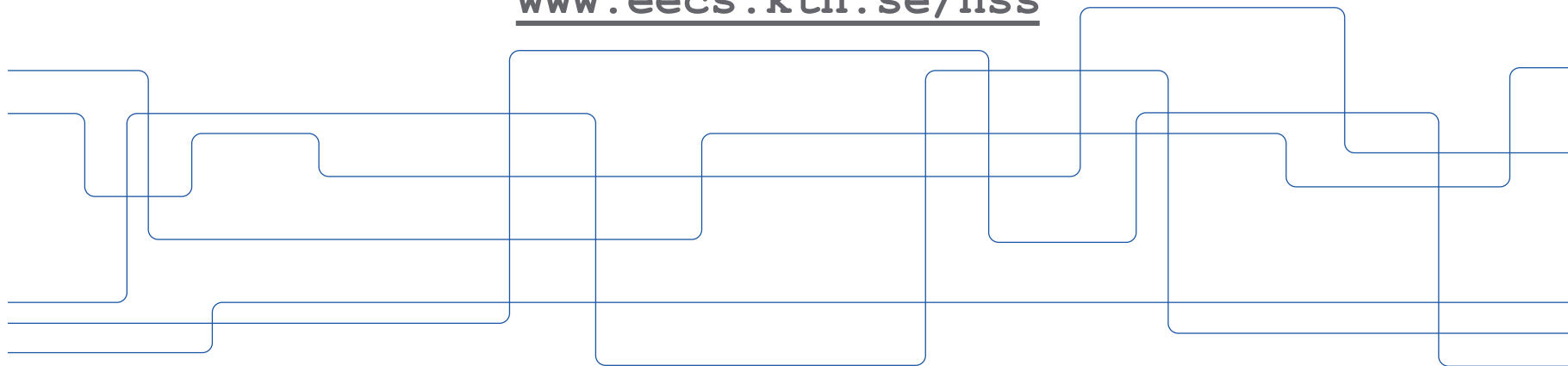
Discussion leads

Zahra Alimadadi &
Marco Casagrande

Secure Sensing

Panos Papadimitratos

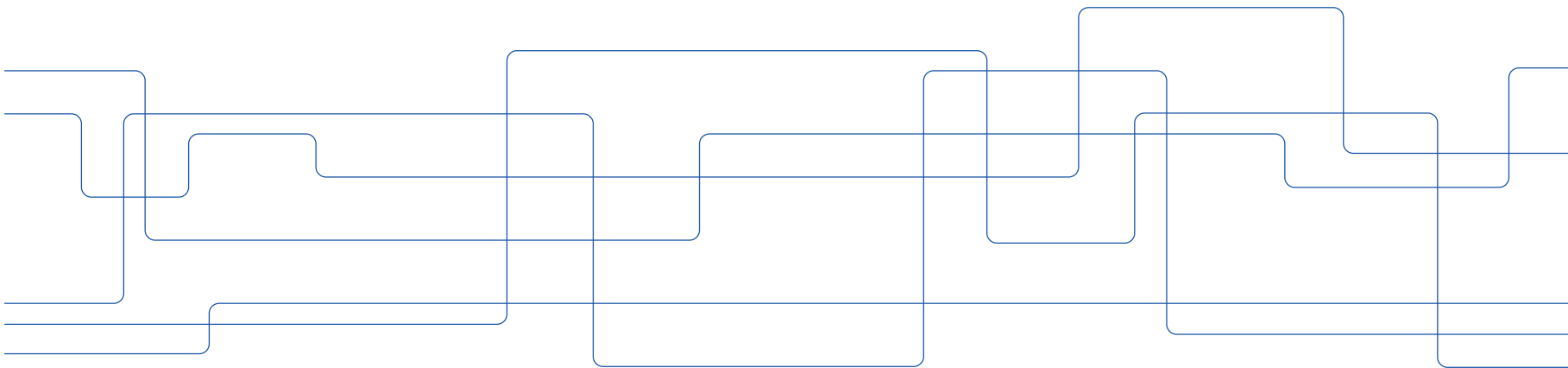
www.eecs.kth.se/nss





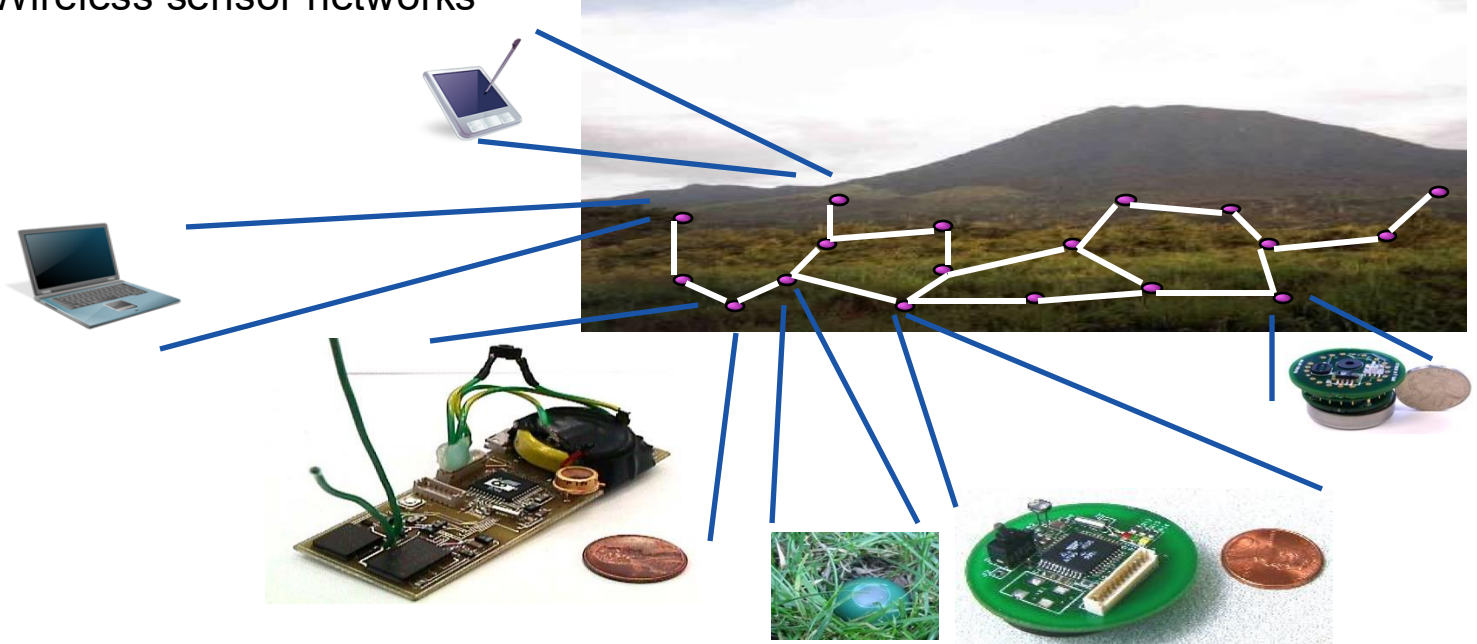
Advanced Networked Systems Security (EP2510/FEP3300)

Introduction



Sensing & Applications

- Wireless sensor networks



Node photos: XBow

Sensing & Applications (cont'd)



Sensing & Applications (cont'd)

Vehicular Communication Systems



Illustration: C2C-CC

Urban Sensing Systems

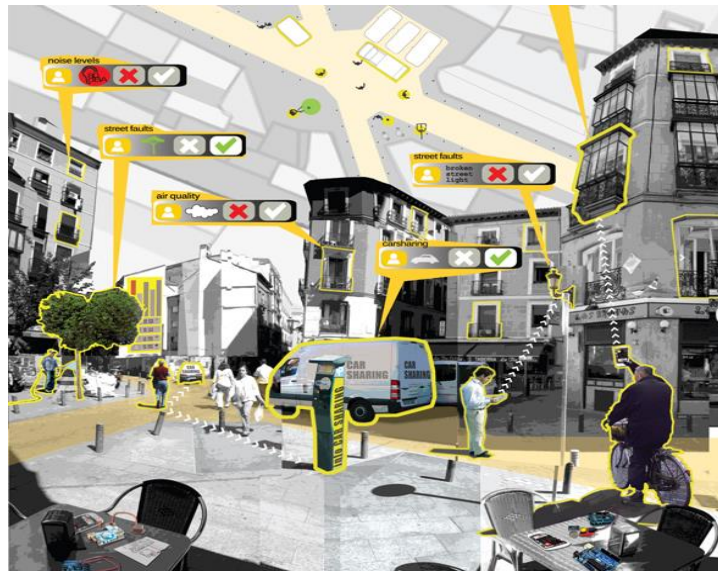


Illustration: complexitys.com

Sensing & Applications (cont'd)

Participatory Sensing



Src: CitiSense : Pollution Monitoring Application
<http://inhabitat.com/>

Human as a Sensor



Src:
<https://www.powerreviews.com/insights/power-of-reviews-survey-2021/>



Src: Sboing4Real - 2022

Gullmarsplan T-bana - 10:54 AM

What's the temperature like?

Freezing cold ☐

Feels chilly ☐

Just right ☐

A little warm ☐

Too hot ☐

Gullmarsplan T-bana - 10:54 AM

How crowded is it on board?

Not crowded
Lots of seats ☐

Not too crowded
Some seats available ☐

Crowded
Limited seating and standing ☐

Very crowded
Limited standing ☐

At capacity
Not taking passengers ☐

Gullmarsplan T-bana - 10:54 AM

Is there security monitoring on board?

Security Guard ☐

Security Camera ☐

Helpline ☐

What accessibility features are available?

Staffed assistance ☐

Accessible entrance ☐

Accessible seating ☐

Accessible stop-button ☐

Hi-visible LED ☐

Submit

Human as a Sensor

Src: nicepng.com, Icons by authors on freeicons.io

Sensing & Applications (cont'd)

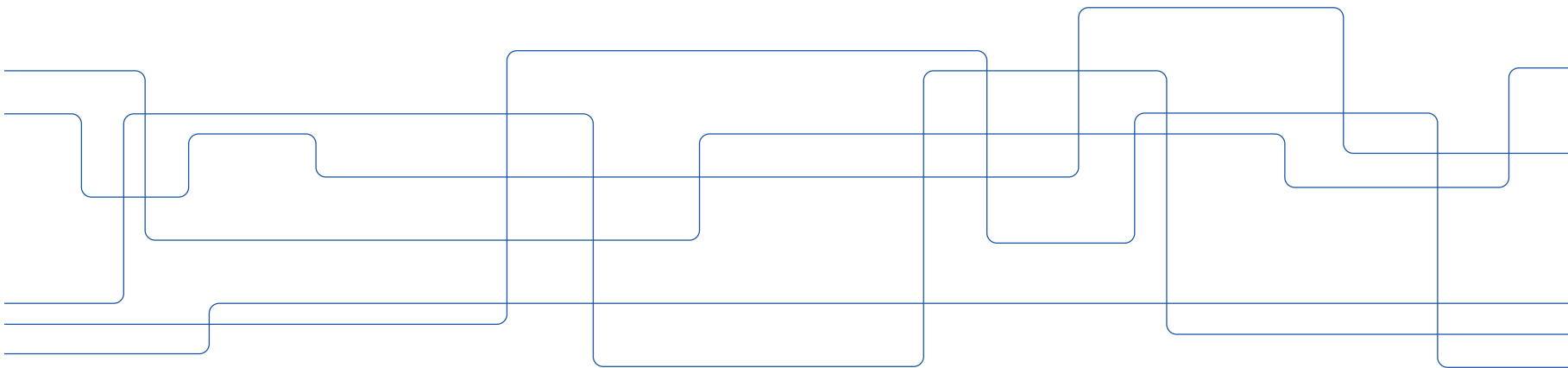


Src: Huang, Haosheng & Gartner, Georg & Krisp, Jukka & Martin, Raubal & Van de Weghe, Nico. (2018). Location based services: ongoing evolution and research agenda. Journal of Location Based Services. 12. 1-31. 10.1080/17489725.2018.1508763.



Advanced Networked Systems Security (EP2510/FEP3300)

Attacks



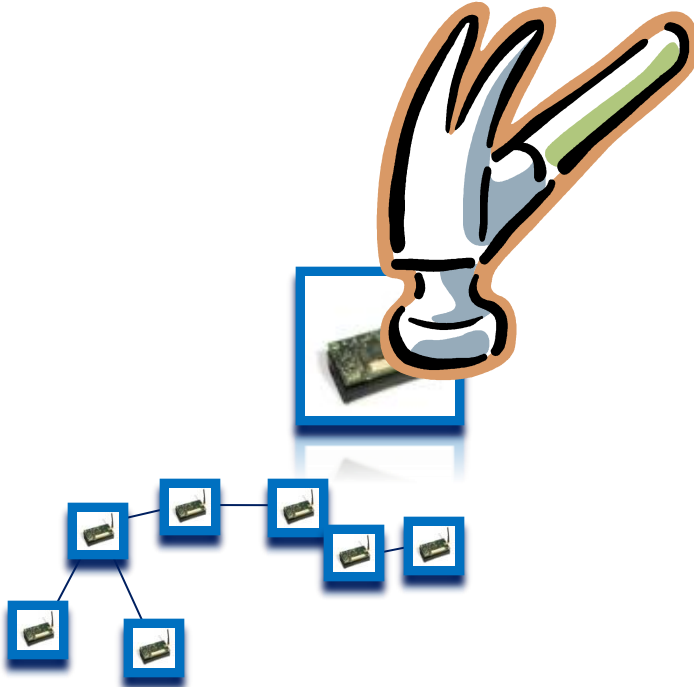
Attack(er) types and motivation

- Client: Users, devices, sensors
 - Active (inject, drop, falsify)
 - > Selfish, e.g., maximize profit and minimize own risk
 - > Malicious, harm/pollute system, arbitrary behaviour
 - > Sybil, impersonate one or more identities
 - Passive
 - > Eavesdropping, gain information
- Server/system: Infrastructure, system components
 - Monitoring: collecting data about clients, tracking
 - Falsification: increasing value of campaign
 - Exploitation/unfairness: make participants do more than what is agreed
 - Semi-honest/honest but curious: correct execution, information kept/inferred
- Outsiders
 - Disrupt (clogging DoS, jamming)

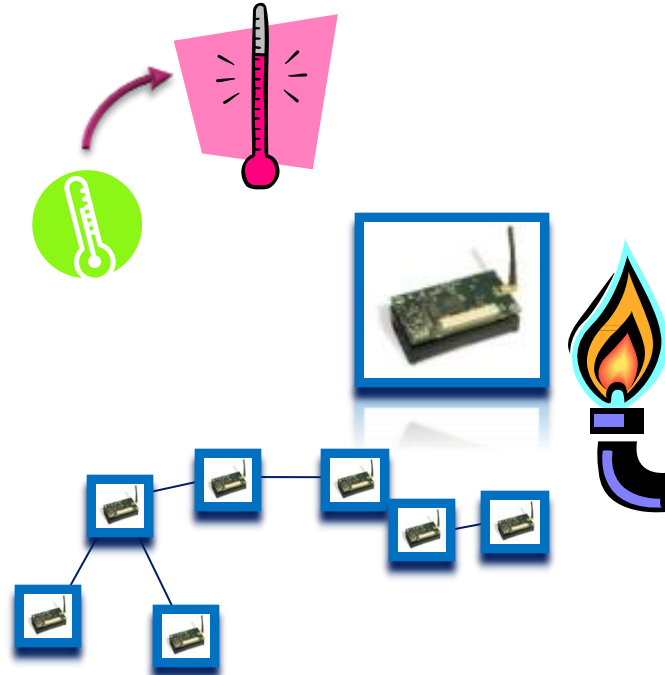


Attacks (cont'd)

- Physical destruction

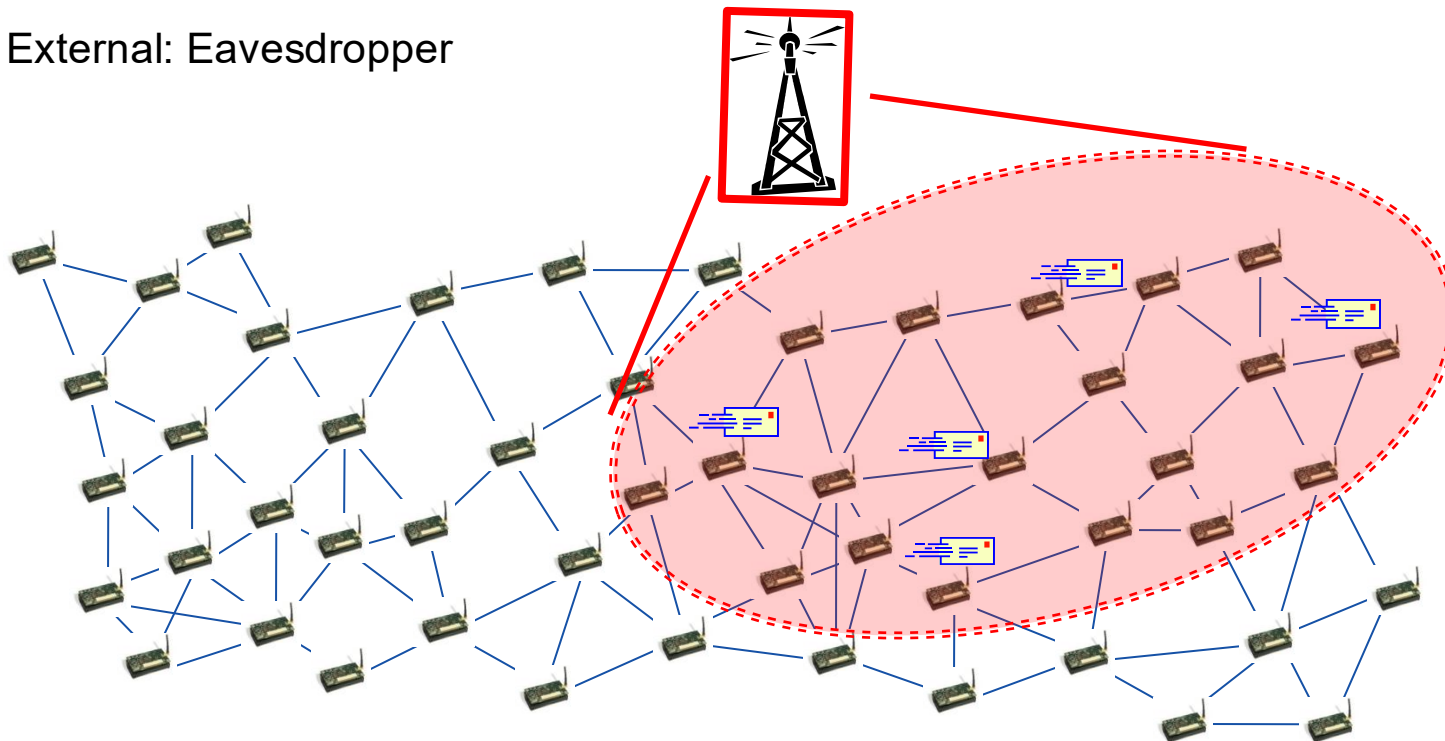


- Input controlling attack



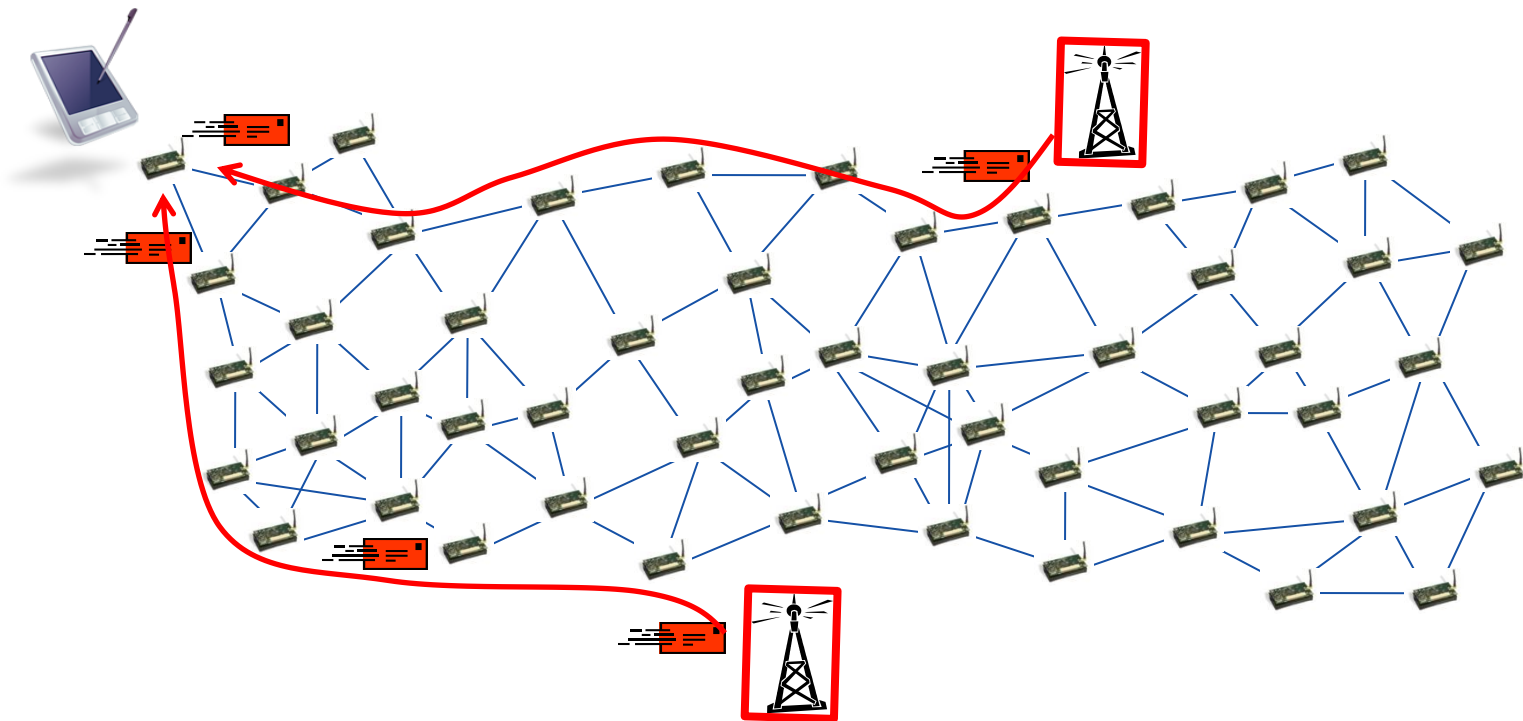
Attacks

- External: Eavesdropper



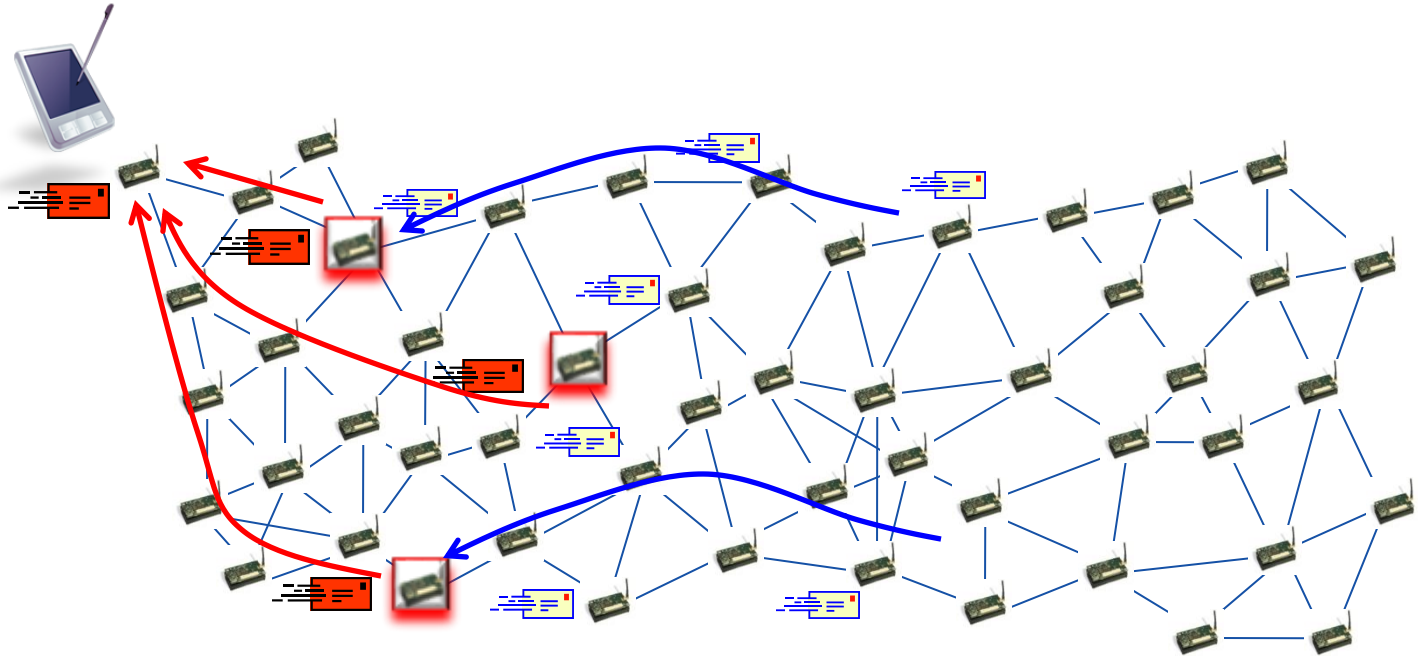
Attacks (cont'd)

- External: Message/measurement injection



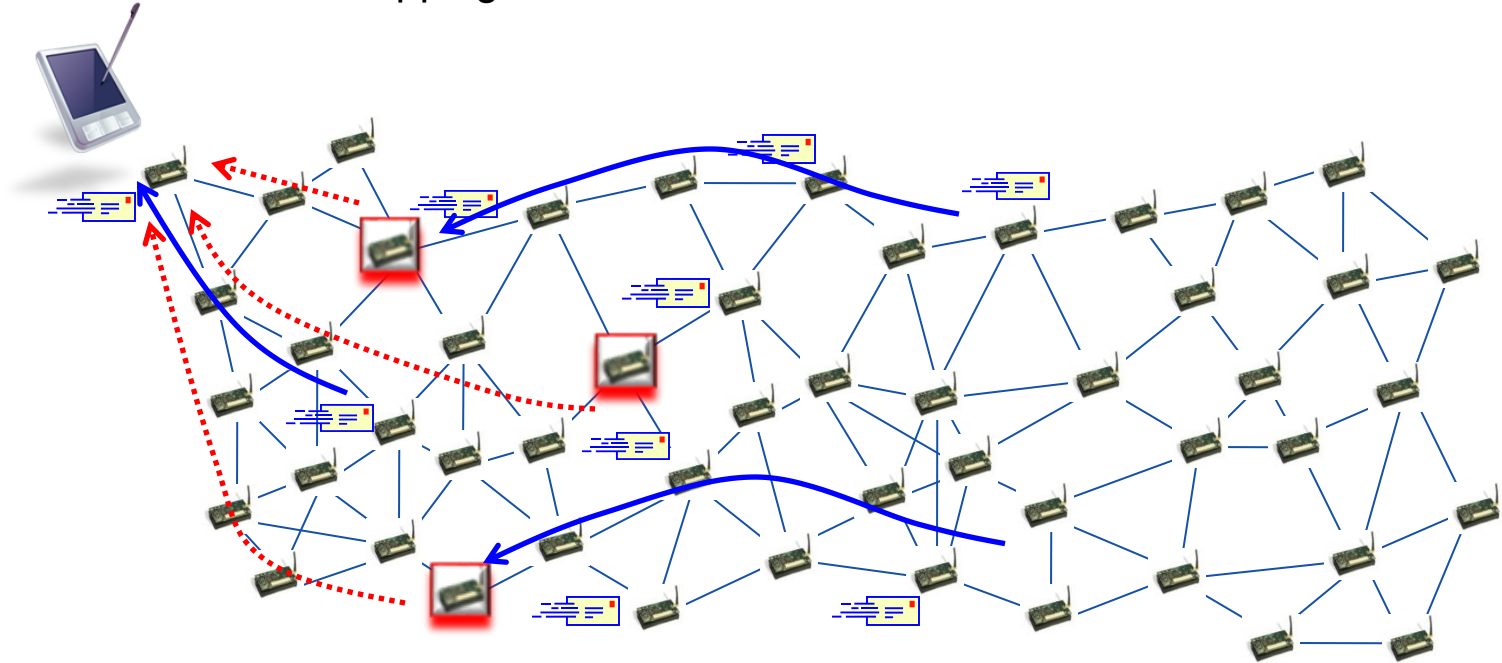
Attacks (cont'd)

- Internal: Packet modification



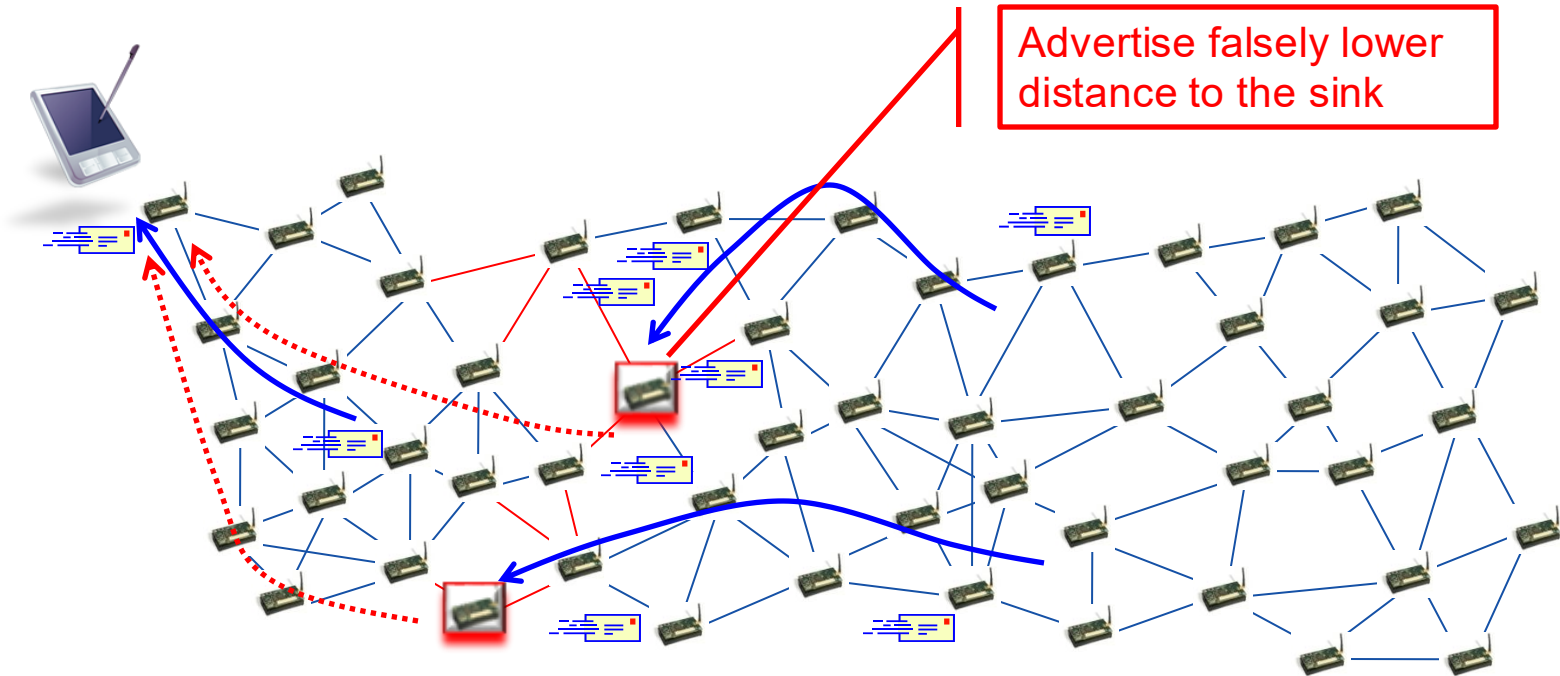
Attacks (cont'd)

- Internal: Packet dropping



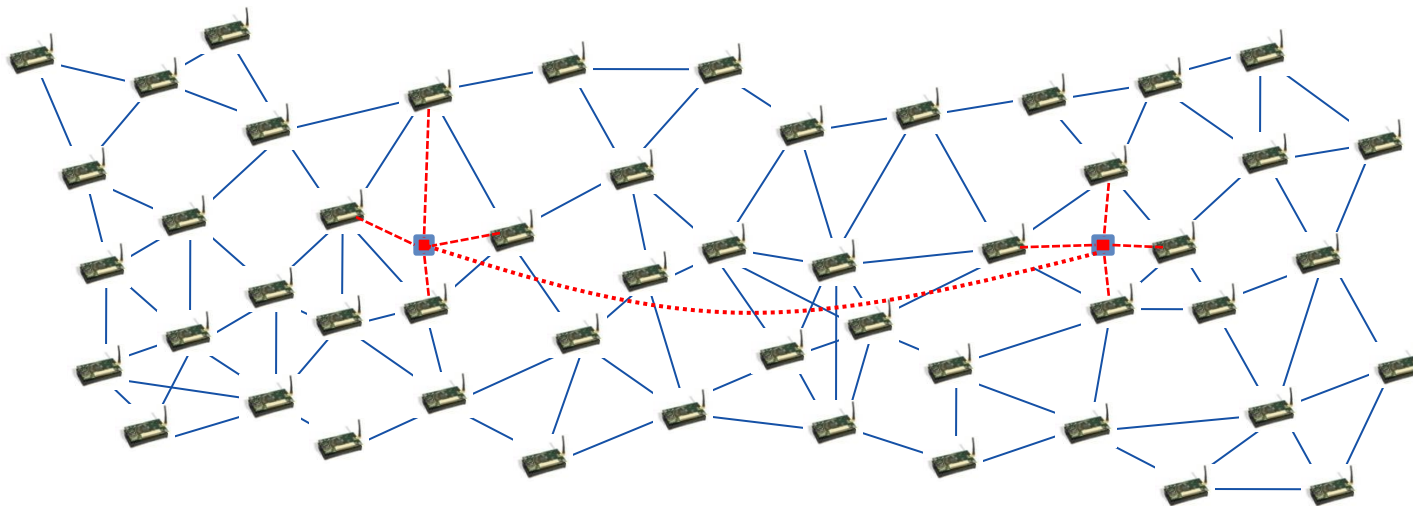
Attacks (cont'd)

- Internal: Packet dropping



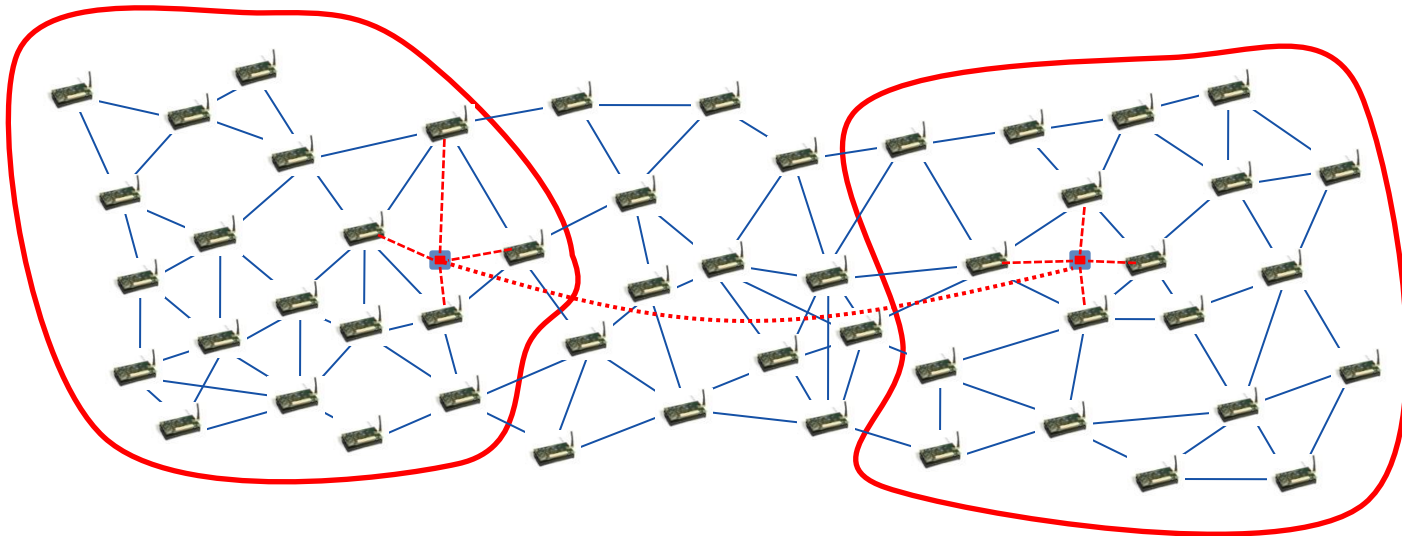
Attacks (cont'd)

- External: Attacking neighbor discovery
 - Create fictitious links
 - Wormhole attack : adversaries 'transport' traffic



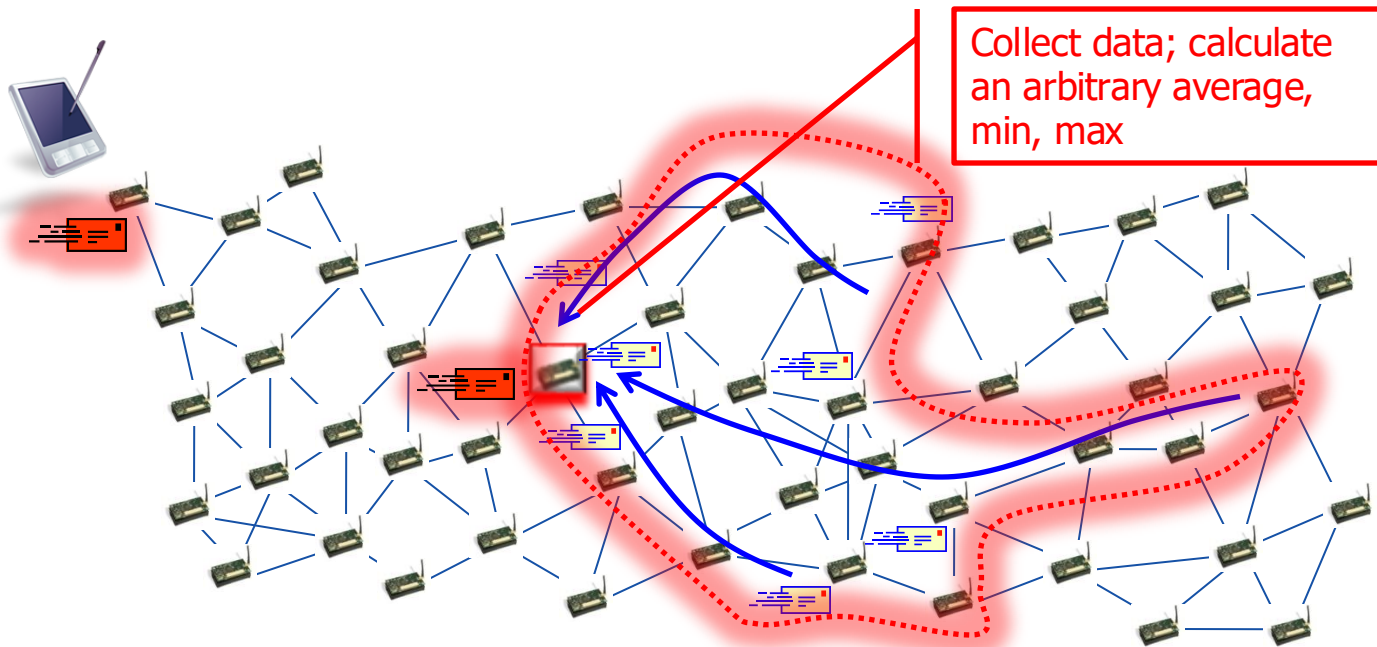
Attacks (cont'd)

- External: Attacking neighbor discovery
 - Create fictitious links
 - Wormhole attack : adversaries 'transport' traffic



Attacks (cont'd)

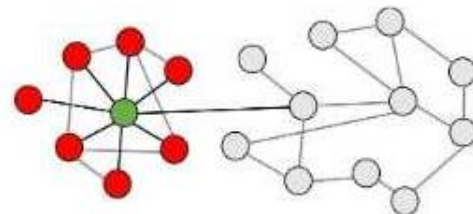
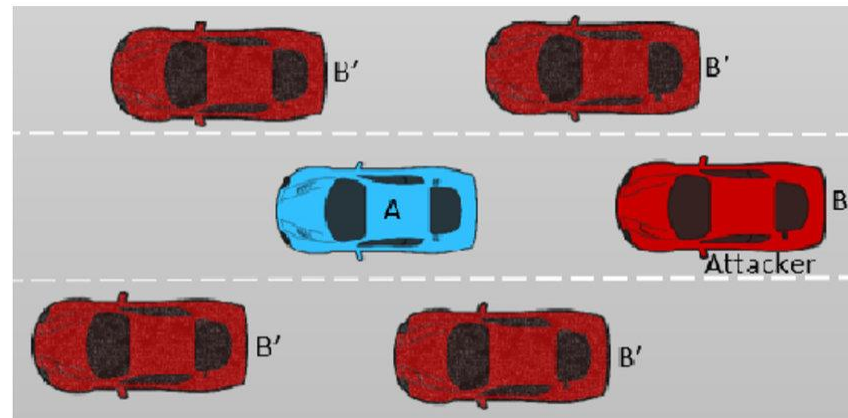
- Internal: Control/corrupt aggregation



Attacks (cont'd)

Sybil behavior, sybil-enabled attacks

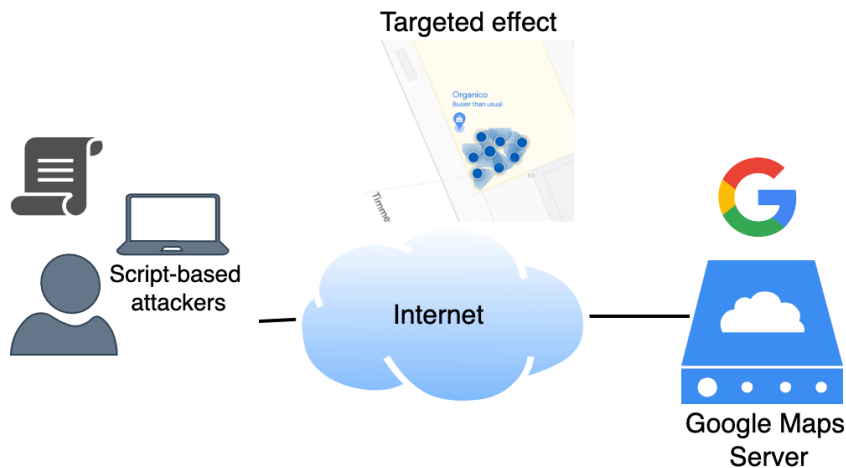
- Use many *identities* to exert influence
 - Create identities/'accounts'/credentials
 - Impersonate/reuse identities
- Execute protocols appearing as multiple participants
 - Submit multiple data measurements
 - Submit multiple 'votes'
 - Act in unison: possibly stronger than collusion
- 'Sybils' can become the majority and control system operation (parts of)



Attacks (cont'd)

Sybil-based Attacks on Google Maps

- Python script to replicate the authentication + data submission
- Falsified/emulated movement
- Multiple submissions concurrently



Algorithm 1 Orchestration of Script-based Place Attacks

Input: User set with credentials: U , target coordinates: loc_{target}

```

1: for all  $s$  such that  $s \in U$  do
2:    $s_{token}, s_{expiry} \leftarrow obtain\_access\_token(s_{RT})$ 
3: end for
4:
5: while True do
6:   for all  $s$  such that  $s \in U$  do
7:     if  $s_{expiry} - t_{now} \leq ONE\_SECOND$  then
8:        $s_{token}, s_{expiry} \leftarrow obtain\_access\_token(s_{RT})$ 
9:     end if
10:    repeat
11:       $x_{geo} \leftarrow create\_xgeo(loc_{target})$ 
12:       $isSubmitted \leftarrow submit(s_{token}, x_{geo})$ 
13:    until  $\neg isSubmitted$ 
14:  end for
15:   $wait(1)$ 
16: end while

```

Cihan Eryonucu, and Panagiotis Papadimitratos. "Sybil-Based Attacks on Google Maps or How to Forge the Image of City Life." ACM WiSec'22 2022



Attacks (cont'd)

Sybil-based Attacks on Google Maps

- Responsible experiments
 - Real Point of Interest (POI)
 - No abuse of account policy
 - Small bump
- Technical advantages
 - Highly efficient, 25 MB memory, 5% CPU usage
 - Scalable
 - Parallizeable

Popular times Fridays ▾



LIVE

Not busy



Popular times Fridays ▾



LIVE

Not busy

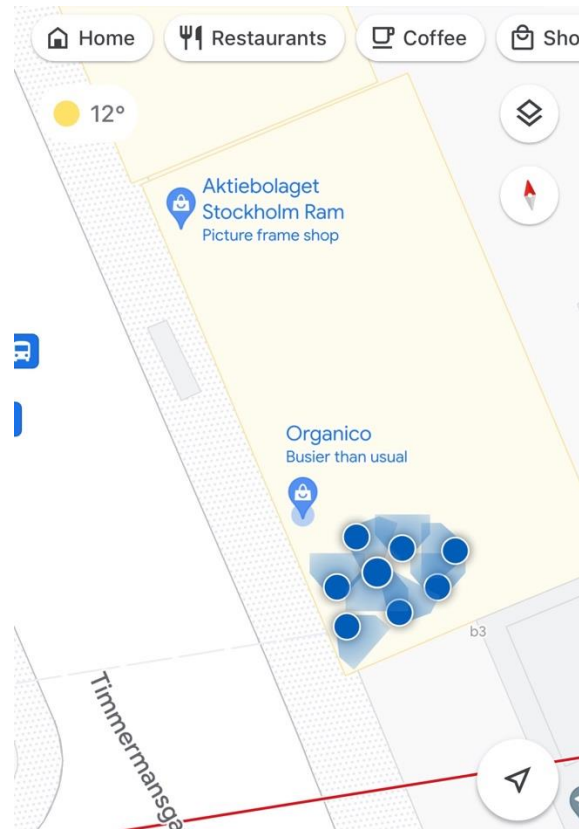




Attacks (cont'd)

Sybil-based Attacks on Google Maps

- Instead of scripts, emulators can submit data
 - More usable
 - > *Easier to configure*
 - Open the app
 - Set the location
 - > *Programmable*
 - Repeat for more emulators
 - Attack: fake popularity

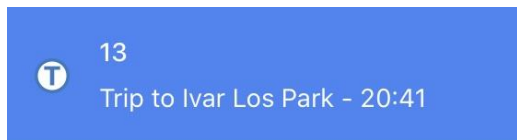




Attacks (cont'd)

Sybil-based Attacks on Google Maps

- Simulate movement to obtain crowdedness forms
 - App fooled by the fake movement <10 seconds
 - Movement is aligned with departure times
 - Auto-clickers to configure the route
- Weak security
 - No emulator checks
 - > *i.e., IMEI number 0*
 - Identical emulators
 - Simultaneous form acquisition with the same account
- Manual attacks are possible
- Responsible action: do not submit form



How crowded was the metro when you got on at **T-Centralen**?

 Many empty seats

 Few empty seats

 Standing room only

 Cramped standing room

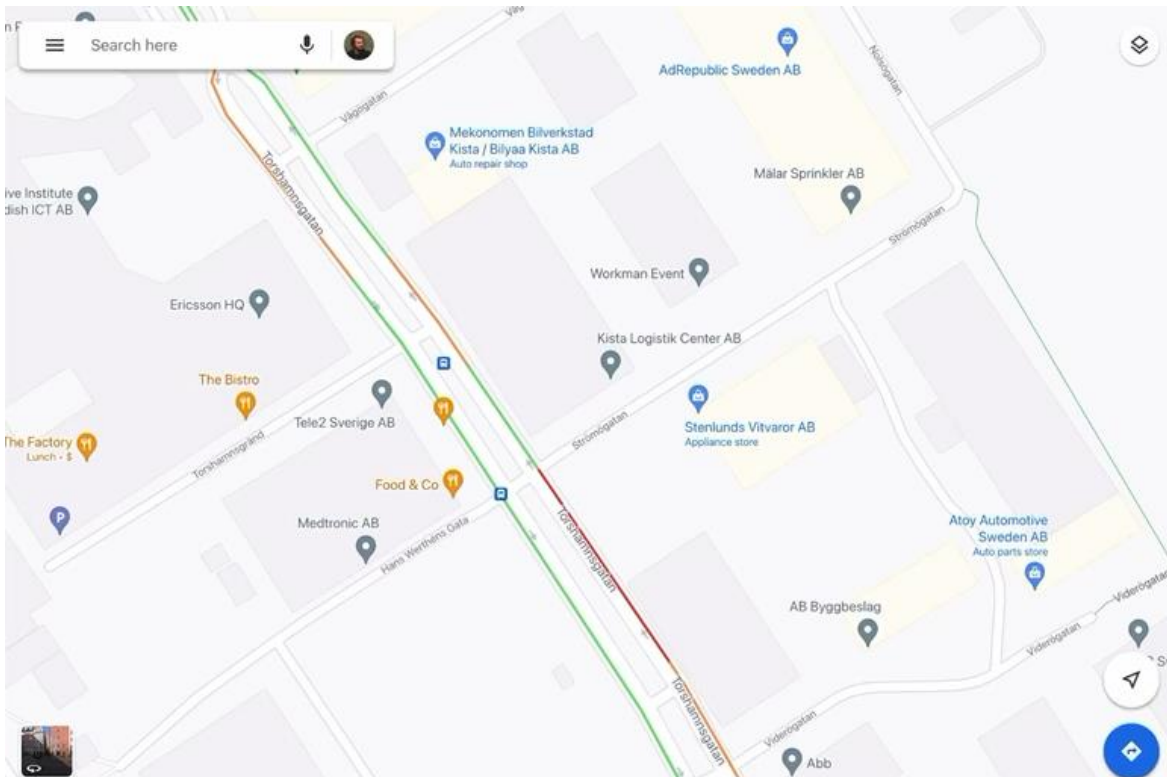
 Not sure

 Didn't take this trip

Attacks (cont'd)

Sybil-based Attacks on Google Maps

- Fake traffic jams
- Simulate a *platoon*
 - 12 slow-moving cars
 - Simultaneous movement
- Road sections turn from green to yellow to red
- Results with 12 emulators
- Responsible action:
Limited impact,
isolated test site





Attacks (cont'd)

Sybil-based Attacks on Google Maps

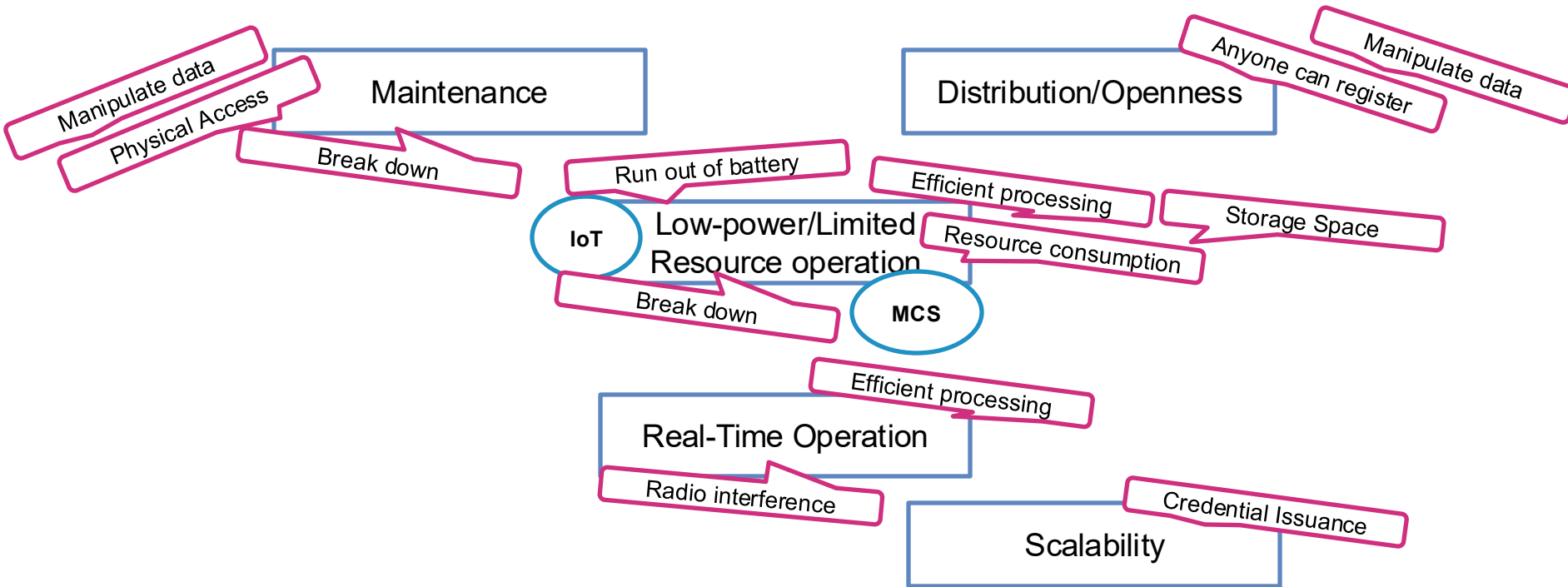
- Most attacks are easy to mount
- Stronger attacker by scaling up
- Mitigation
 - Timed and non-overlapping tokens
 - Data verification schemes
 - Emulator detectors

Attacks (cont'd)

Challenges

Static sensor

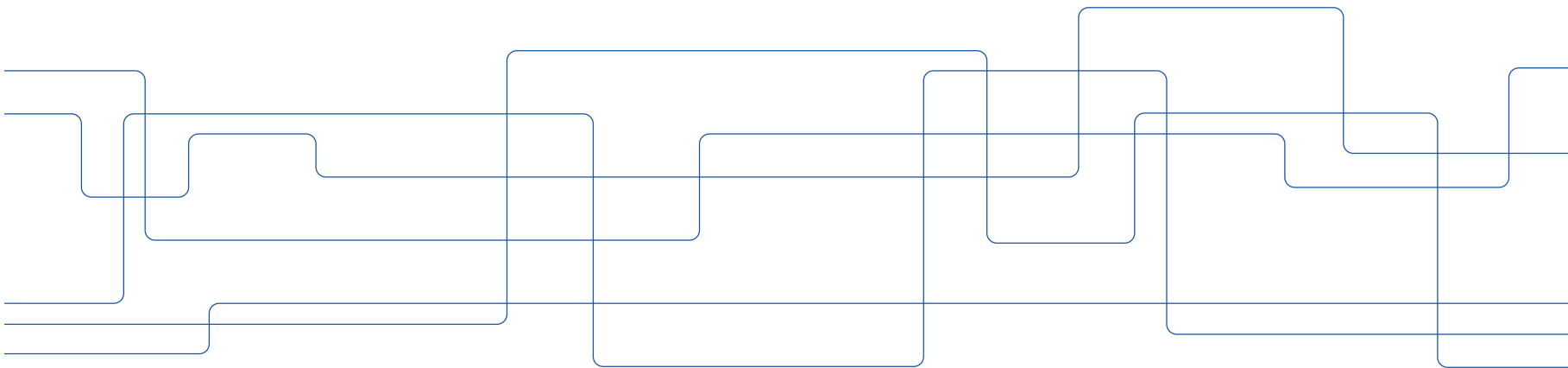
Mobile sensor



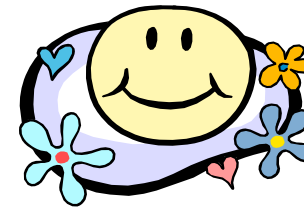
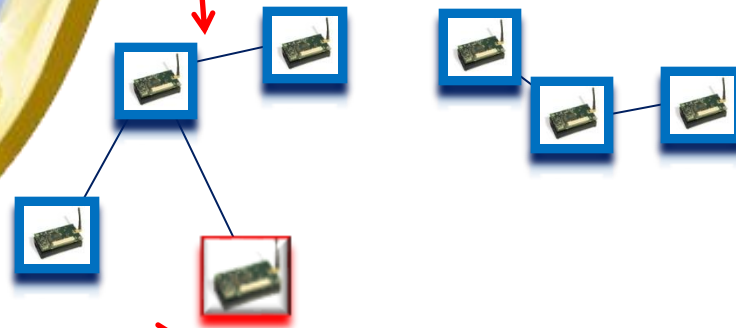


Advanced Networked Systems Security (EP2510/FEP3300)

Safeguarding sensing & applications



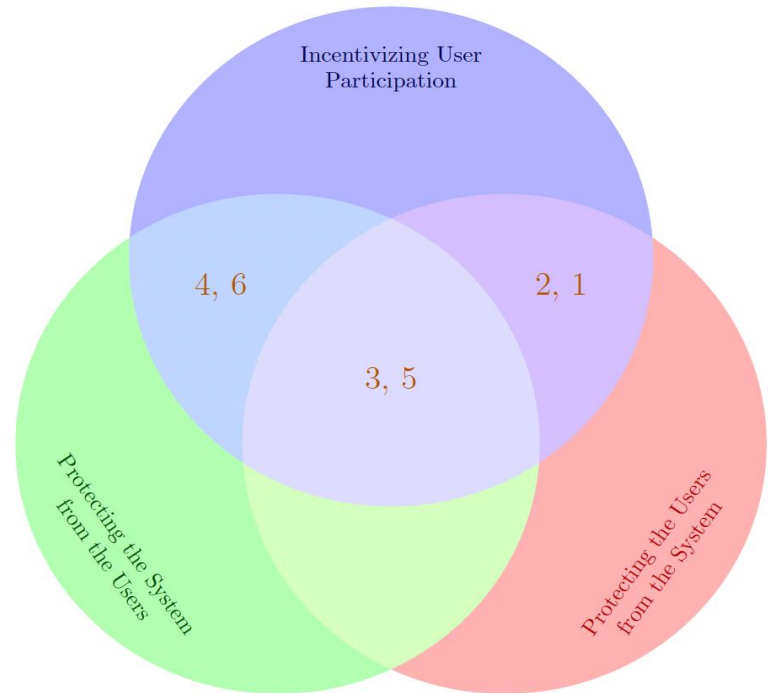
- Prevent
- Manage



Safeguarding sensing

Security and Privacy Requirements

1. Privacy (-preserving participation)
 - Anonymity (conditional)
 - > *Unobservability*
 - Protection against inference attacks
 - Unlinkability
2. Privacy-preserving, resilient incentive mechanisms and fairness
3. Communication Integrity
Confidentiality and Authentication
4. Authorization and Access Control
5. Data-Centric Trust
6. Accountability



T. Giannetsos, S. Gisdakis, and P. Papadimitratos, "Trustworthy People-Centric Sensing: Privacy, Security and User Incentives Road-Map," in IEEE IFIP Mediterranean Ad Hoc Networking Workshop (IEEE IFIP MedHocNet), 2014.



Safeguarding sensing (cont'd)

- Cryptography
 - Symmetric/Asymmetric Key Crypto
 - > *Key Management (distribution, discovery, establishment, update)*
 - Anonymous Credentials
- Data Validation
 - Machine learning
 - Game theory
 - Belief theory



Key Management

- Public key
 - Simple, effective
 - > *Broadcast authentication*
 - > *Distribution of new keys and insertion of new nodes is straightforward*
 - Expensive
 - > *Even though several implementations fitting small footprint platform are available now*
- Symmetric key approaches predominant
 - Caution: Symmetric keys grow fast in numbers
 - > *For a network of N nodes, N^2 symmetric keys*
 - > *Maybe less, depending on the utilized secure(d) protocols*

L. Eschenauer and V. D. Gligor, “ A key-management scheme for distributed sensor networks”. In Proceedings of the 9th ACM conference on Computer and communications security (CCS '02), 2002.

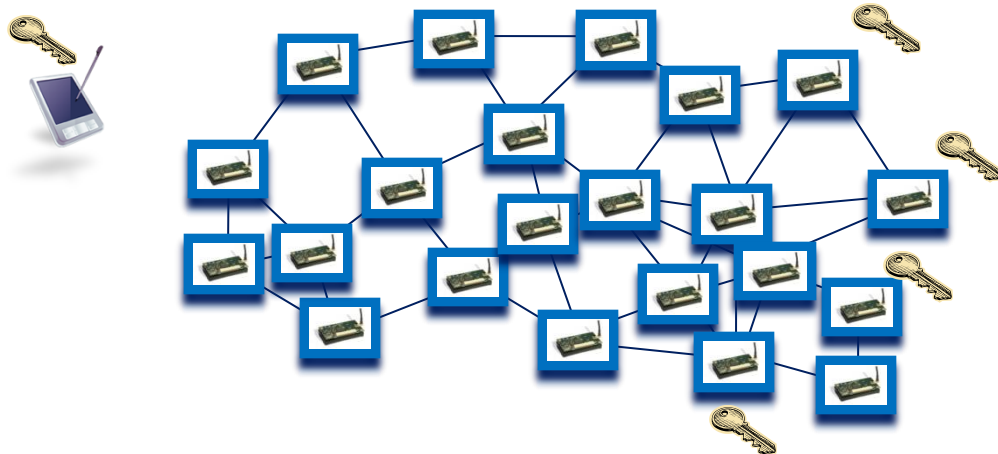


Key Management (cont'd)

- How many keys, shared by which nodes, why, how, for how long?
- Key Distribution (or Pre-Distribution)
- Key Discovery
- Key Establishment
- Key Update

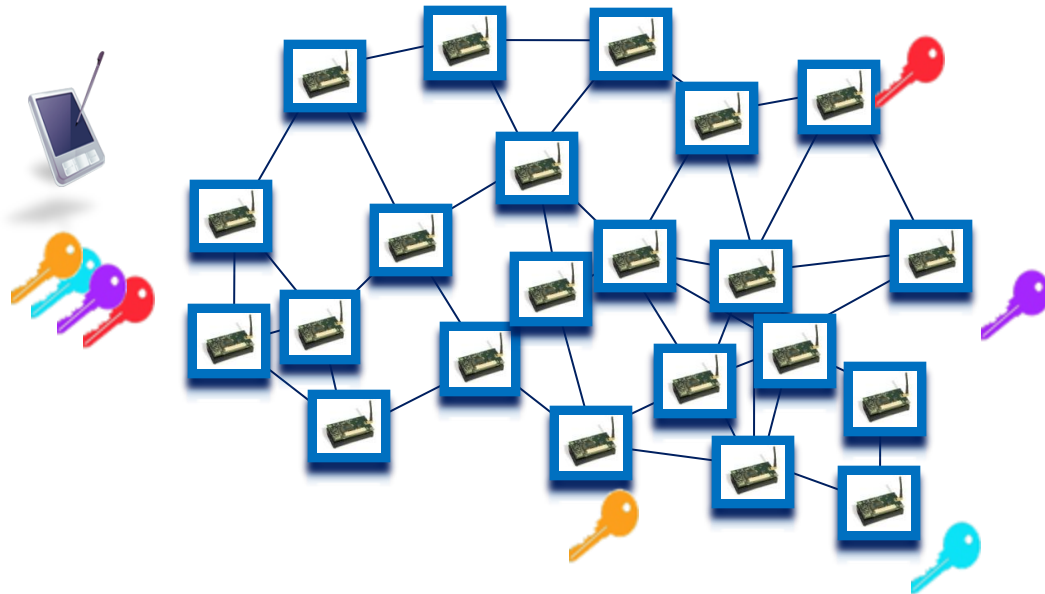
Key Management (cont'd)

- One key for all
 - Low security
 - Low storage
 - Efficient broadcast authentication
 - No resilience to compromise
 - Easy to add new nodes



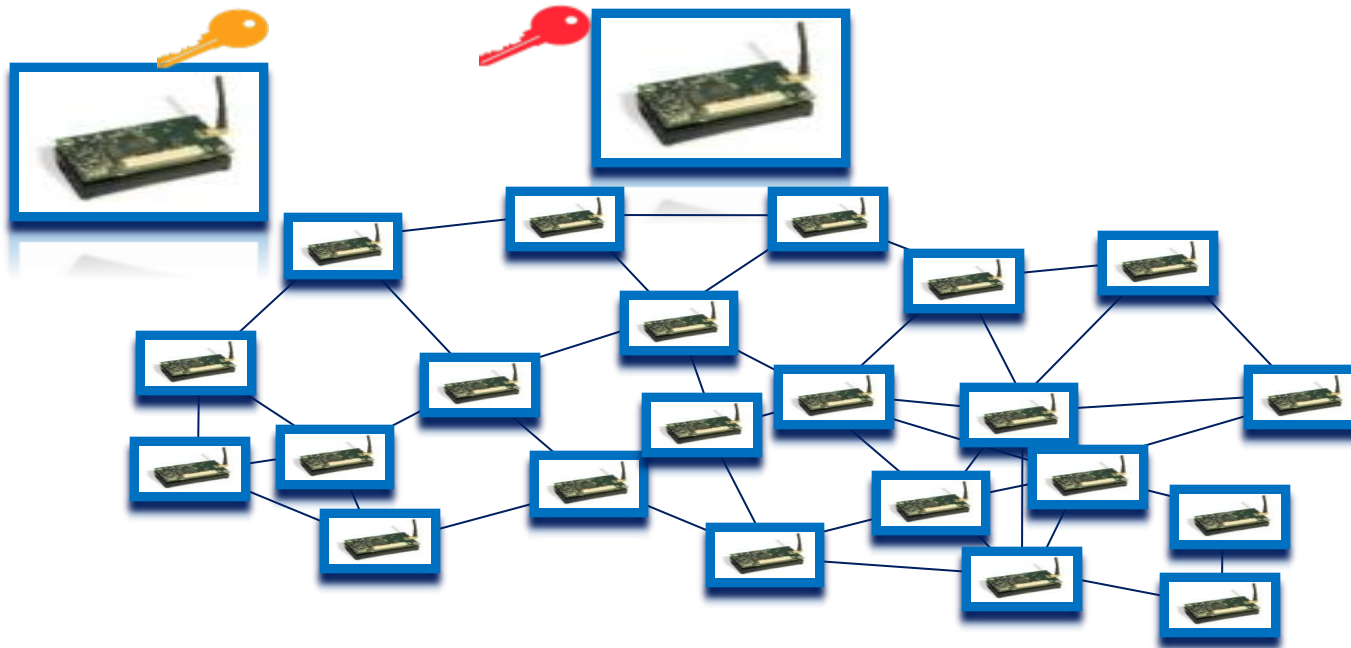
Key Management (cont'd)

- One key per node (shared with the sink)
 - high storage (n keys)
 - inefficient broadcast authentication
 - resilience to node compromise
 - not expensive to add new nodes



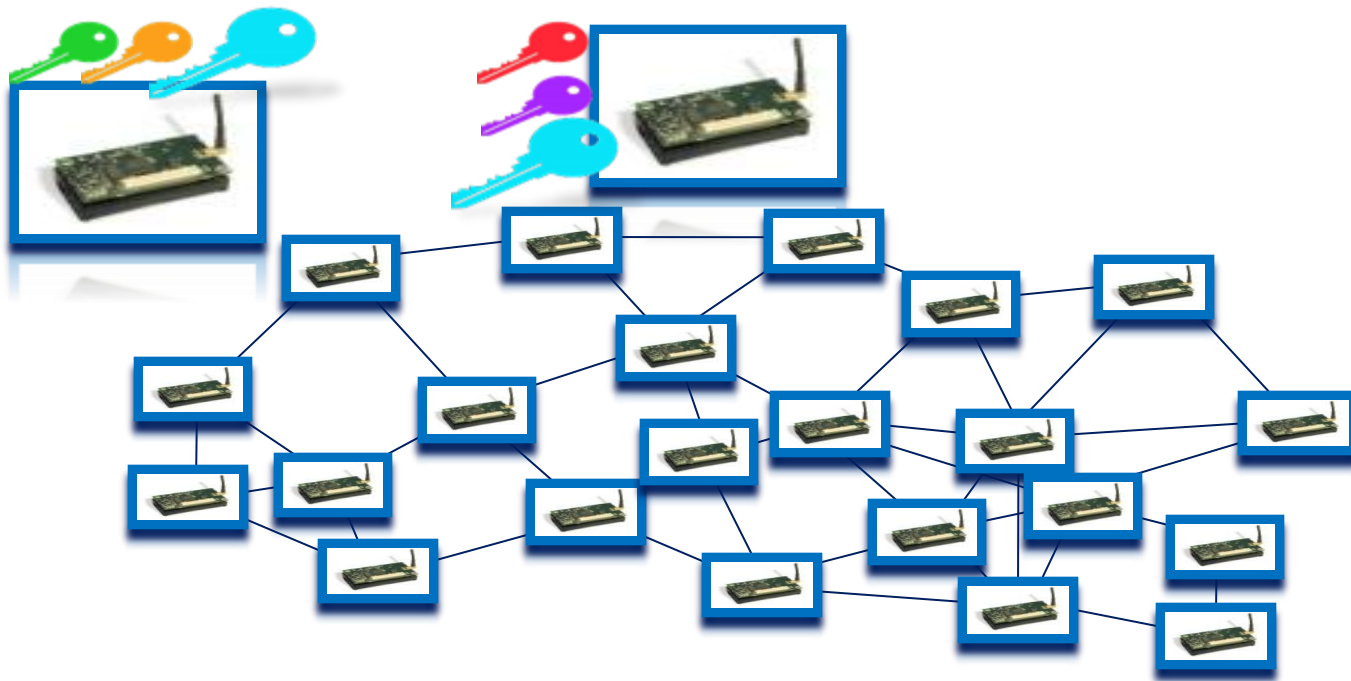
Key Management (cont'd)

- What about node-to-node authentication?



Key Management (cont'd)

- What about node-to-node authentication?





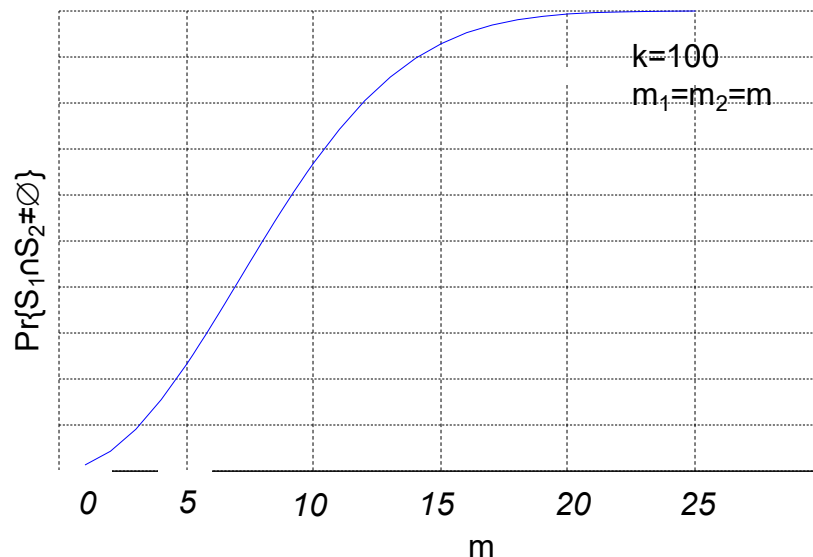
Key Management (cont'd)

- Initialization
 - Large pool S of unique keys are picked at random
 - > *The whole pool is known only to the authority*
 - Each node is preloaded with m keys selected randomly from S
- Key discovery phase
 - Which neighbors do I share a key with?
 - > *each sensor node broadcasts a key identifier list to one-hop neighborhood (more than one pair may share the same key)*
- Key establishment
 - Establish a key with a neighbor through another neighbor
 - > *if two sensor nodes still do not share a key*
- How many keys should each node get?

Key Management (cont'd)

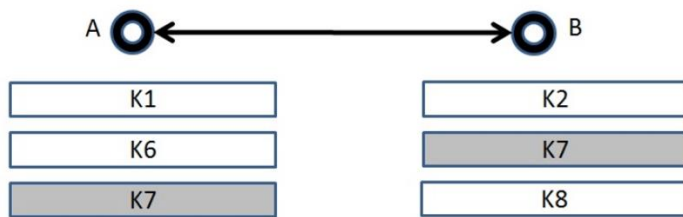
- Dimension (number of keys) the *key ring* of each node (m_1, m_2) and the *key pool* (k), so that probability that any two nodes share at least one key gets the sought value

$$\Pr\{S_1 \cap S_2 \neq \emptyset\} = 1 - \frac{(k - m_1)!(k - m_2)!}{k!(k - m_1 - m_2)!}$$

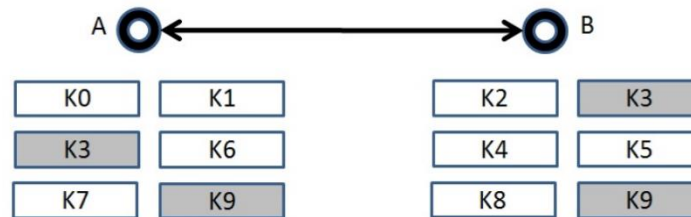


Key Management (cont'd)

Random Key pre-Distribution



RKD scheme for A and B that discover a shared key of size L



Key splitting scheme for nodes A and B that discover two shared key parts of size $L/2$

M. Ehdaie, N. Alexiou, M. Ahmadian, M. R. Aref, and P. Papadimitratos, "Key Splitting for Random Key Distribution Schemes," IEEE Workshop on Secure Network Protocols (IEEE NPSec), in conjunction with IEEE , 2012

Data confidentiality & parasitic adversaries

- Assumption 1: Individual sensor measurements to be collected
 - Scientific measurements, industrial field and inventory data, health patient data
- Assumption 2: Data of interest
 - For a network of N nodes, the user (sink) needs to collect data:
 - > From δN nodes, where $0 < \delta \leq 1$
 - > For a period $T = kt_0$, where t_0 is one measurement period
- Assumption 3: Sensor nodes know a path to the sink

P. Papadimitratos, J. Luo, and J.-P. Hubaux, “Randomized Countermeasure Against Parasitic Adversaries in Wireless Sensor Networks,” *IEEE Journal on Selected Areas in Communications (IEEE JSAC)*, 2010

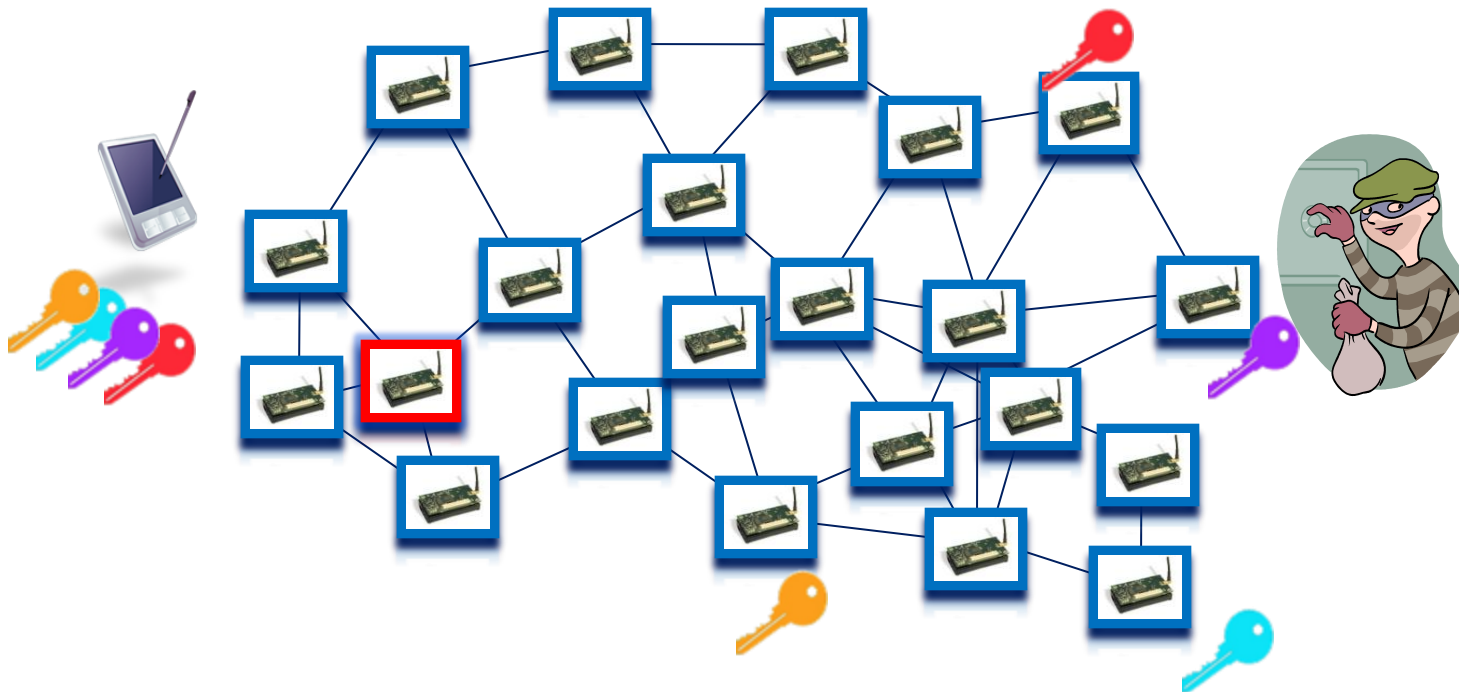


Data confidentiality & parasitic adversaries (cont'd)

- Parasitic adversary
 - No intention to damage the targeted sensor network
 - Obtain (δ, T) sensor measurements with the least expenditure of own resources
 - Can be physically present, at each point in time, only at a much smaller fraction of the area covered by δN sensor nodes
 - Can physically access data stored at sensor nodes and retrieve their cryptographic keys
 - Can be mobile, i.e., compromise different sets of nodes over different time intervals
 - Can compromise in the above-described manner at most one sensor per τ seconds; $\tau \ll T$

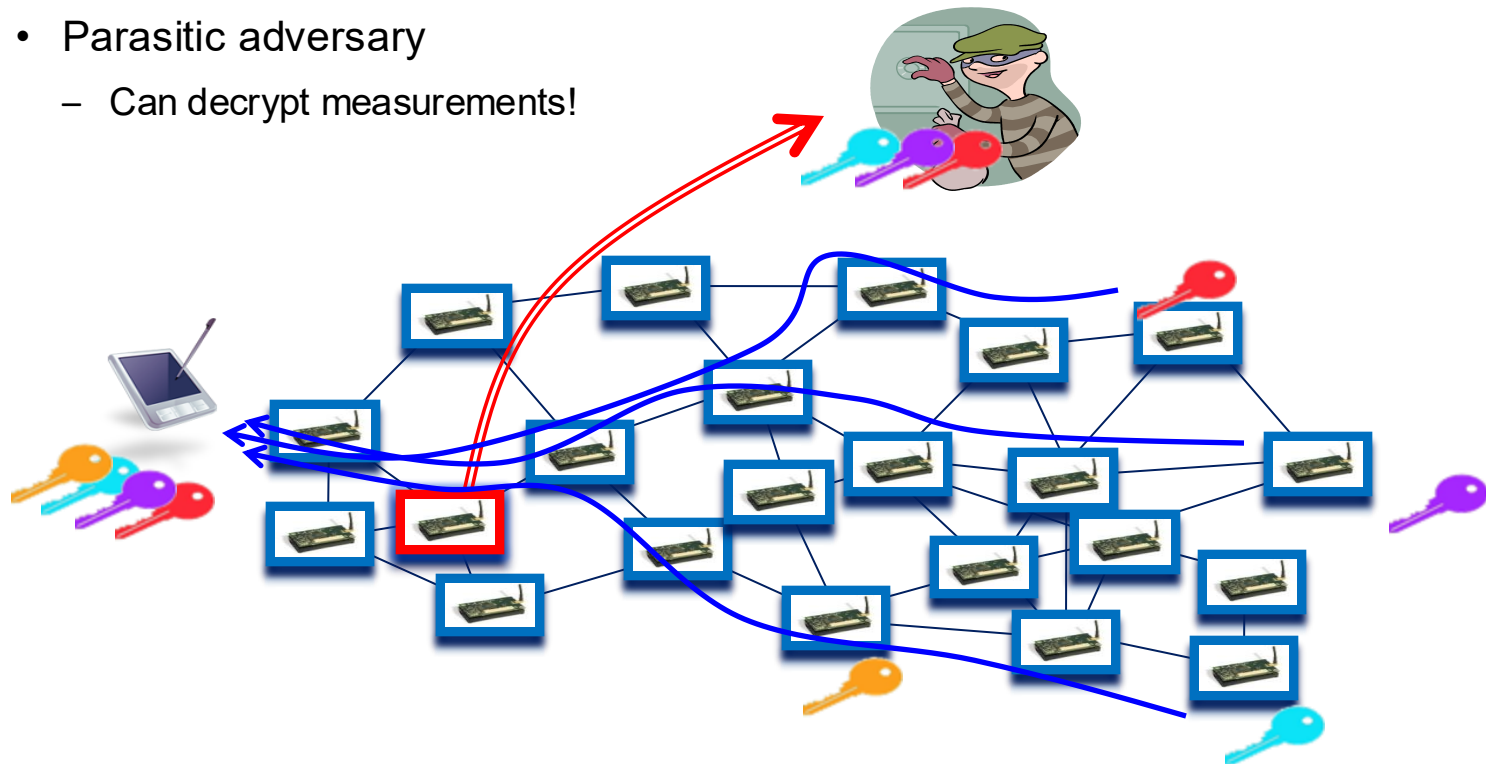
Data confidentiality & parasitic adversaries (cont'd)

- Parasitic adversary



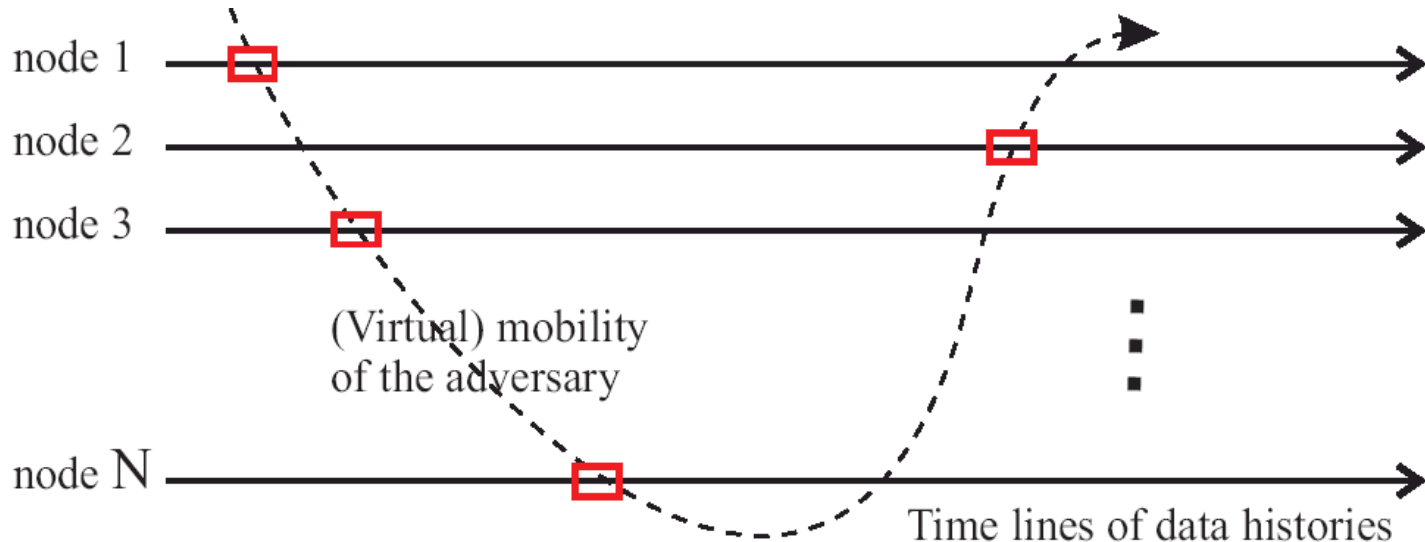
Data confidentiality & parasitic adversaries (cont'd)

- Parasitic adversary
 - Can decrypt measurements!



Data confidentiality & parasitic adversaries (cont'd)

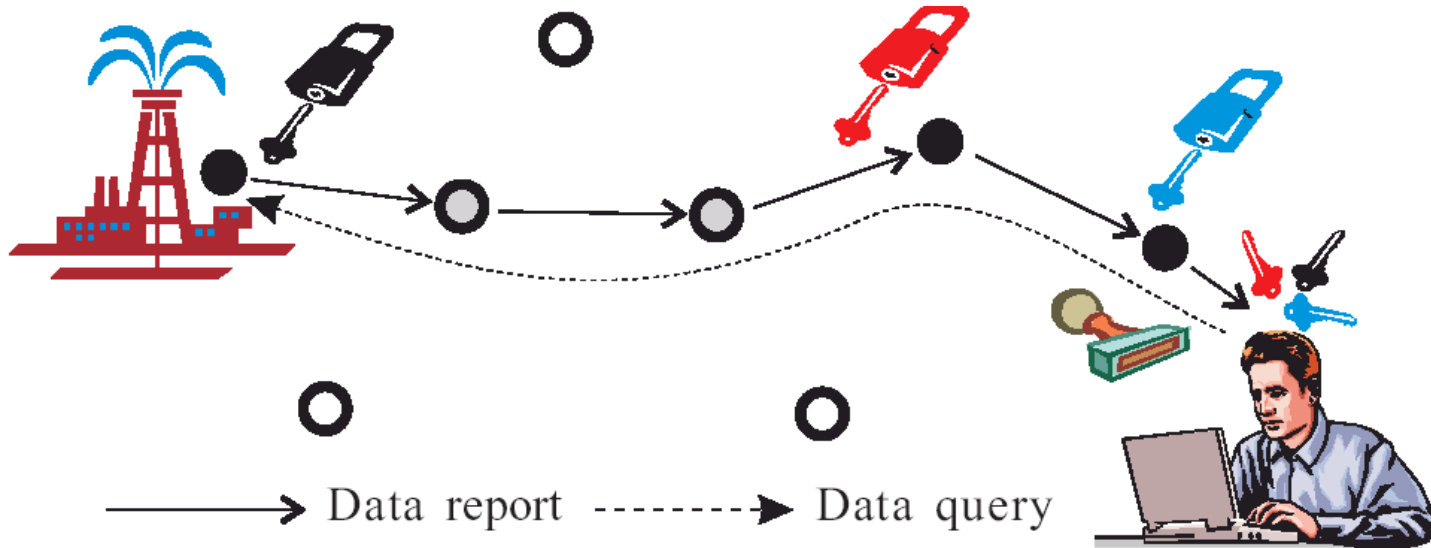
- Realistic
 - Assumption: cannot modify the functionality of nodes



Data confidentiality & parasitic adversaries (cont'd)

En route re-encryption

- Source node always encrypts its data
- Intermediate nodes “flip a coin” and re-encrypt packets with probability q





Data confidentiality & parasitic adversaries (cont'd)

Key Refreshing

- Sensor nodes randomly elect a point in time to execute the key refreshing protocol
 - The time for the protocol execution is drawn randomly around a system specific parameter for the average refresh rate
 - The sensor node generates a new symmetric key
 - It encrypts it with the sink public key
 - It transmits it as if it were a regular data measurement

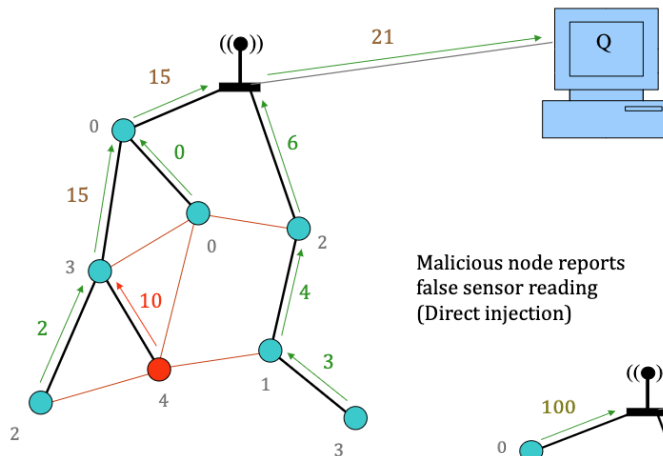
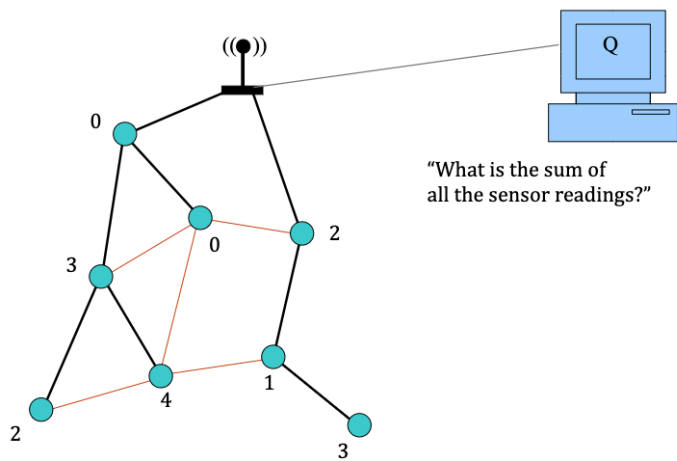


Data confidentiality & parasitic adversaries (cont'd)

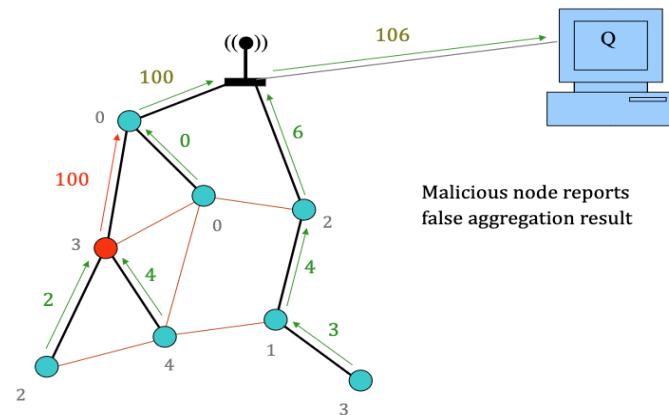
Key Refreshing

- The sink does *not* know which node is physically compromised
 - It can only guess which part of the network the adversary may target
 - It is still clueless among that subset of nodes which are compromised
- The adversary does not know which node is refreshed
 - It could guess if symmetric key transport was used, and
 - > *It were within one hop from the refreshed node*
 - > *It had compromised the entire path from from the source*
- The adversary can always re-read the key of previously compromised nodes

Aggregation result falsification



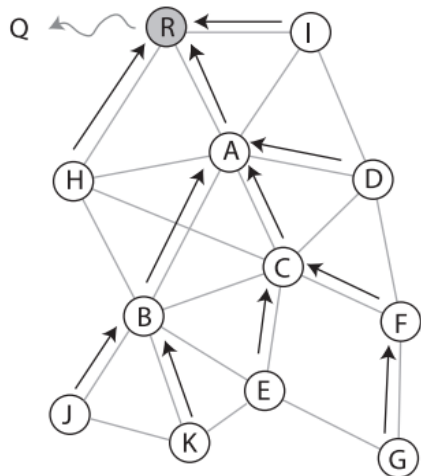
Malicious node reports false sensor reading (Direct injection)



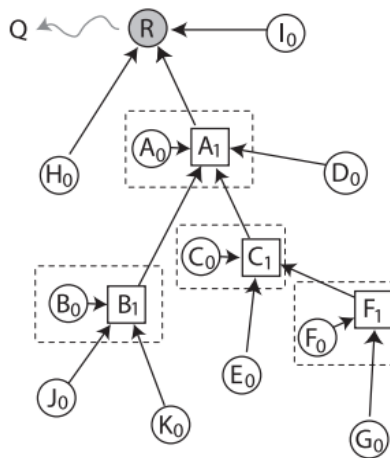
Malicious node reports false aggregation result

H. Chan, A. Perrig, and D. Song, "Secure hierarchical in-network aggregation in sensor networks". In Proceedings of the 13th ACM conference on Computer and communications security (CCS '06). 2006

Aggregation and Naive Commitment Tree in Network Context



(a) Example network graph. Arrows: Aggregation tree. R: Base station. Q: Querier.



$$R = \langle 12, v_R, \bar{v}_R, H[N||12||v_R||\bar{v}_R||H_0||A_1||I_0] \rangle$$

$$A_1 = \langle 9, v_{A_1}, \bar{v}_{A_1}, H[N||9||v_{A_1}||\bar{v}_{A_1}||A_0||B_1||C_1||D_0] \rangle$$

$$C_1 = \langle 4, v_{C_1}, \bar{v}_{C_1}, H[N||4||v_{C_1}||\bar{v}_{C_1}||C_0||E_0||F_1] \rangle$$

$$F_1 = \langle 2, v_{F_1}, \bar{v}_{F_1}, H[N||2||v_{F_1}||\bar{v}_{F_1}||F_0||G_0] \rangle$$

$$G_0 = \langle 1, a_G, r - a_G, G \rangle$$

(b) Naive commitment tree, showing derivations of some of the vertices. For each sensor node X , X_0 is its leaf vertex, while X_1 is the internal vertex representing the aggregate computation at X (if any). On the right we list the labels of the vertices on the path of node G to the root.

Efficient and Robust Secure Aggregation for Sensor Networks

- SHIA detects an attack but then needs to collect individual data
 - No in-network aggregation benefit
- Solution: detect, localize and evict attackers
 - Adversary localizer
 - > Mark nodes that disrupt the data collection
 - > Mark nodes that disrupt the ACK propagation
 - Aggregation Tree Reconstruction
 - > Reform the aggregation tree excluding the marked nodes

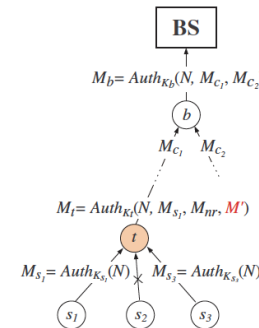
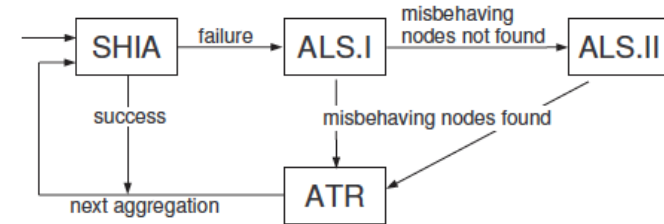


Fig. 2. ALS.I: Hierarchical Collection of Confirmations. Nodes s_1 and s_3 acknowledge, node s_2 does not. Note t is faulty: it encapsulates M_{s1} , and M_{nr} correctly, but modifies the message of s_3 . The collection continues, until node b provides the final confirmation to the BS.

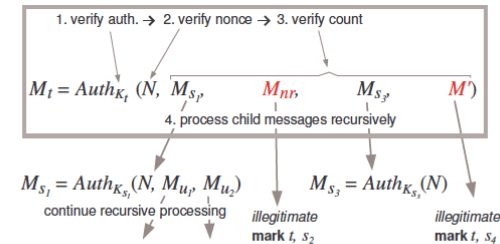
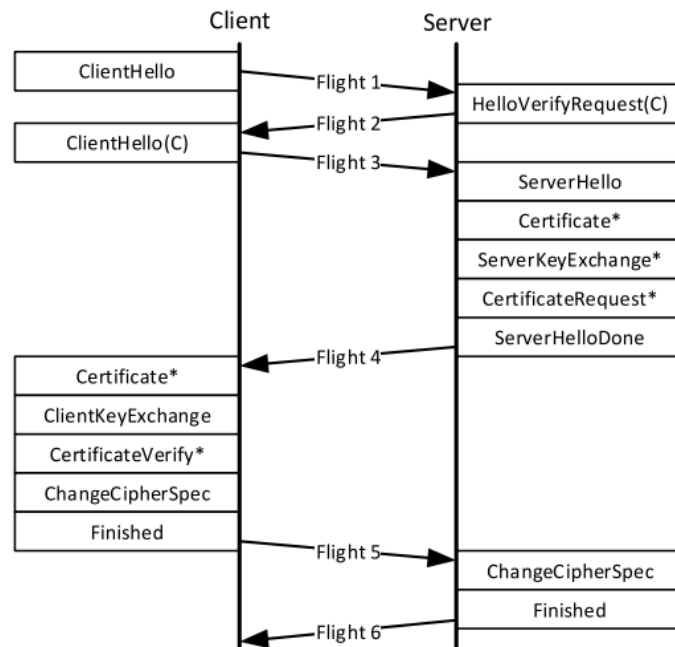
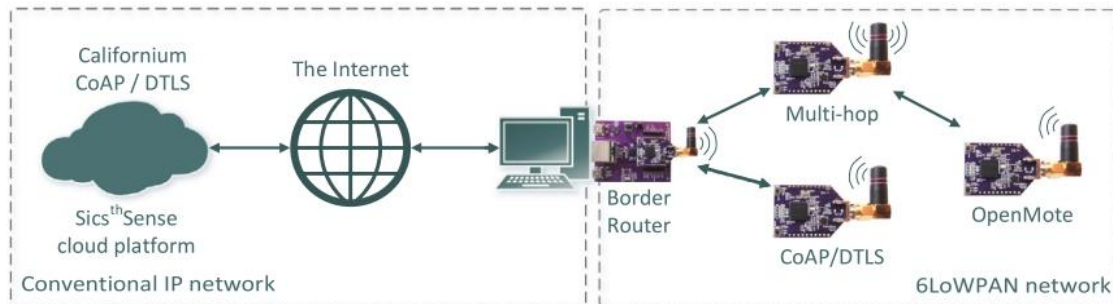


Fig. 3. ALS.I: Recursive Processing of Confirmations: recursive procedure steps, explained in the box, and unfolding of the recursion.

P. Haghani, P. Papadimitratos, M. Poturalski, K. Aberer, and J. Hubaux, "Efficient and Robust Secure Aggregation for Sensor Networks," IEEE Workshop on Secure Network Protocols (IEEE NPSec), in conjunction with IEEE ICNP, Beijing, China, pp. 1-6, October 2007

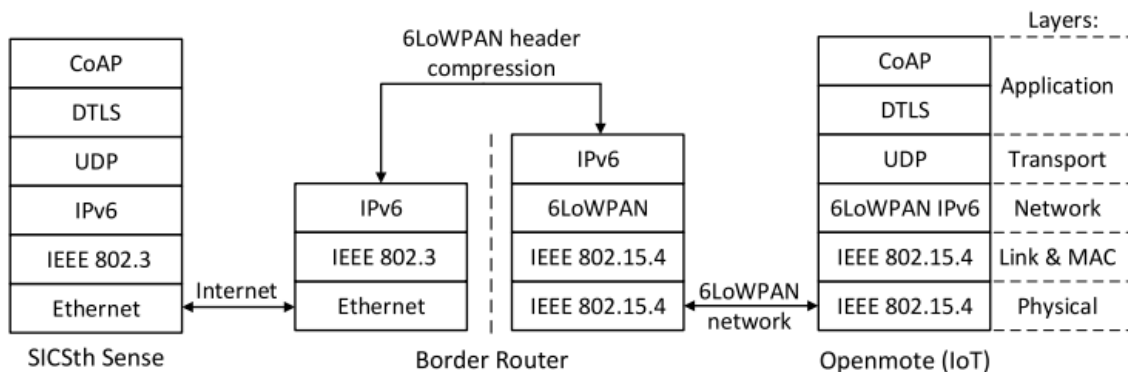
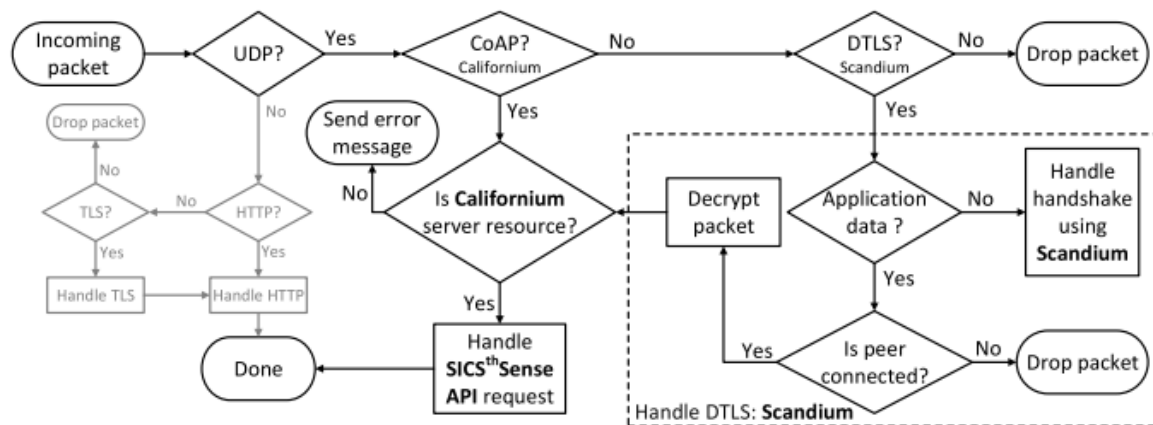
SecureSense

- CoAP as de-facto standard
 - Variant of HTTP, runs over UDP
 - DTLS is mandatory
- E2E secure communication
 - Pre-shared Key (PSK) based
 - Certificate based
 - Raw Public Key (RPK) based



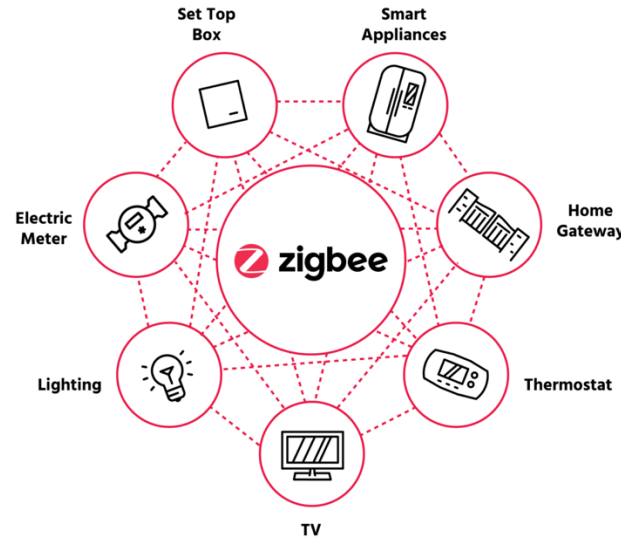
S. Raza, T. Helgason, P. Papadimitratos, and T. Voigt, "SecureSense: End-to-end Secure Communication Architecture for the Cloud-Connected Internet of Things," *Future Generation Computer Systems*, 2017

SecureSense



Zigbee

- Network standard for low-powered, bandwidth and range devices
- Operates as a mesh network
 - Perfect for smart-home setting
 - Stability when a node fails
- Security Design:
 - Open trust
 - Inter-device comm. is encrypted
 - > *E2E encryption*
 - Trust Center
 - > *Optional*
 - > *Decides on policies*
 - > *Key establishments and generations*
 - Two security modes

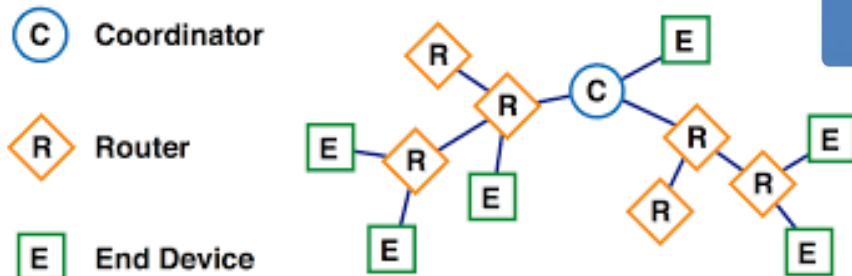
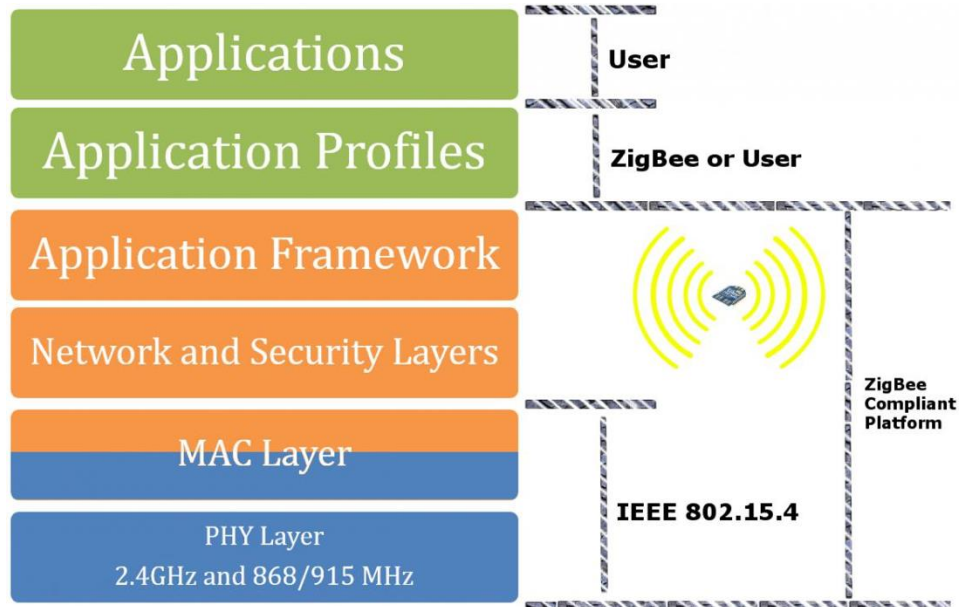


Smart Home

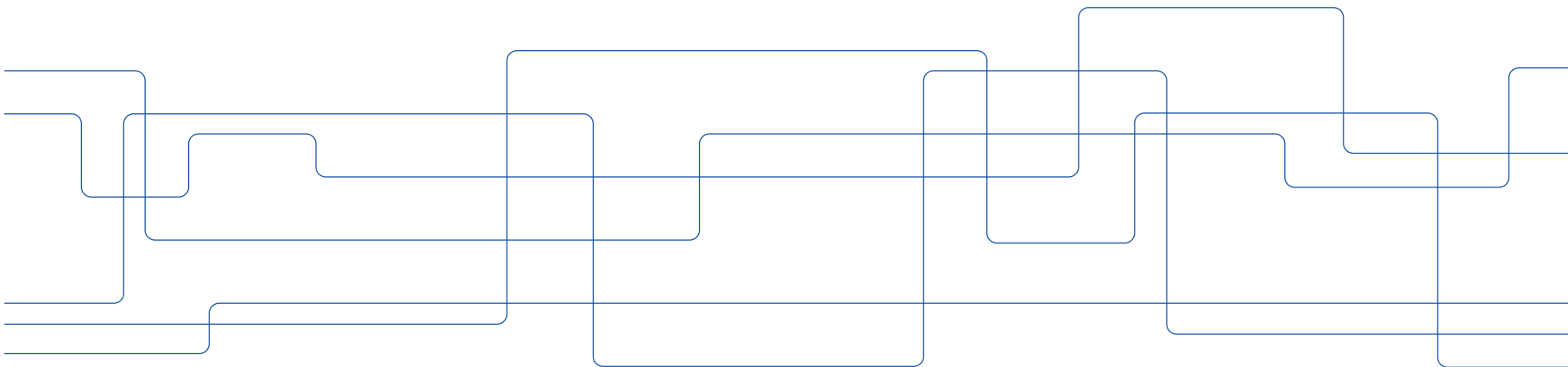


Zigbee Security

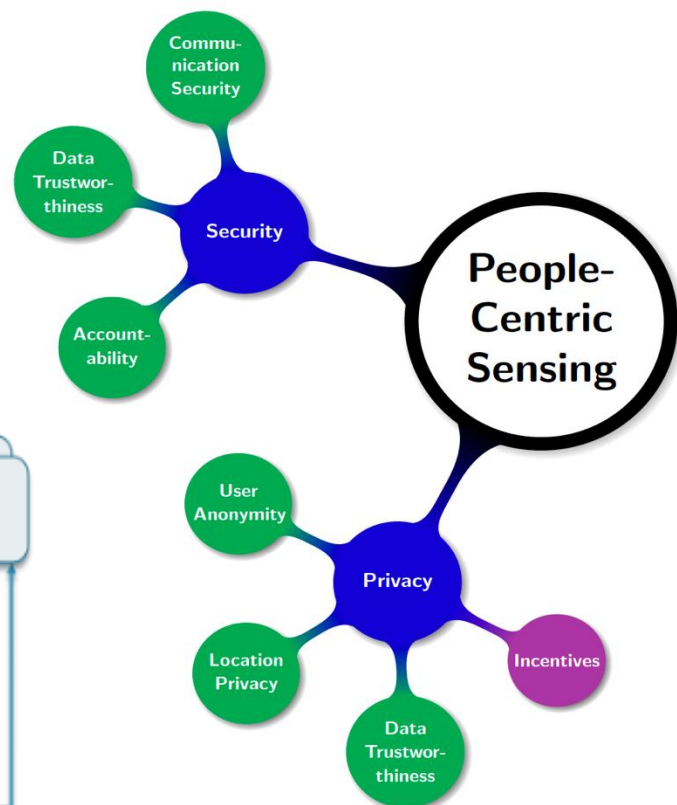
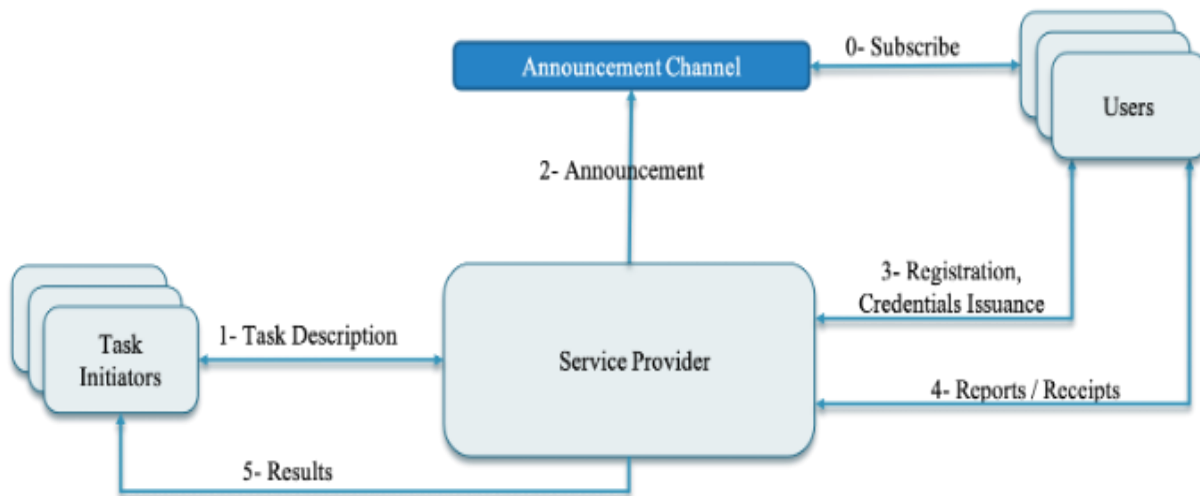
- Distributed Mode
 - Trust center is optional
 - Network key is transmitted when joined
- Centralized Mode
 - Trust center control and coordinates
 - Authentication of devices by TC
 - Maintaining all keys



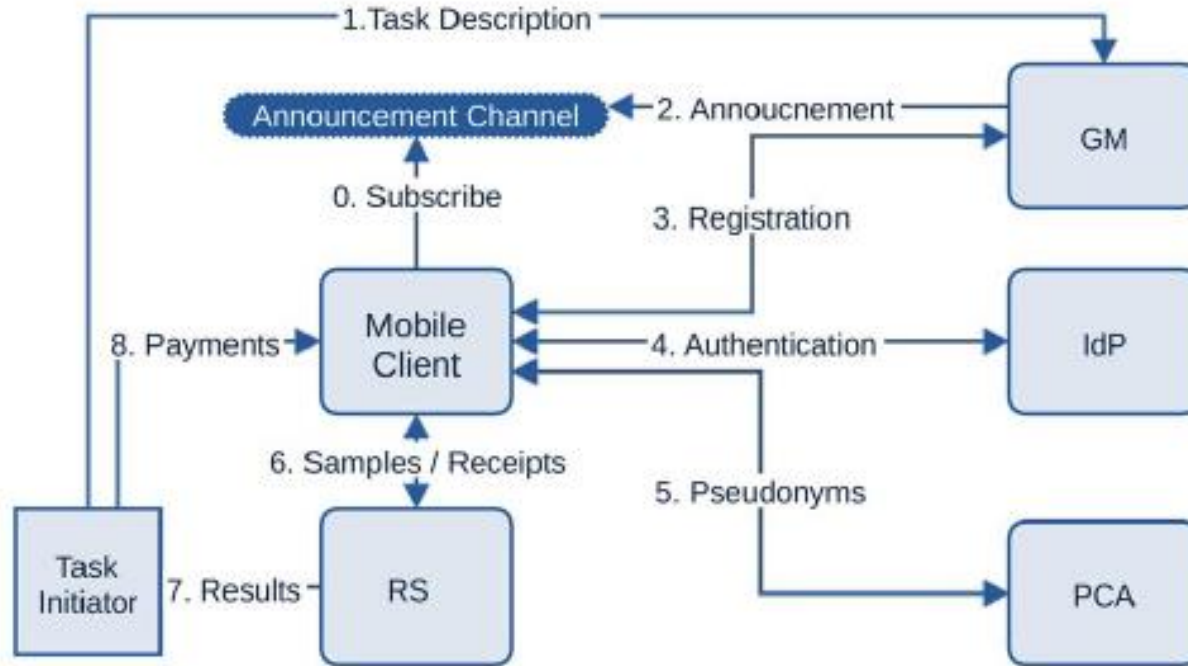
Participatory Sensing



System & Requirements



Participatory Sensing Security Architecture

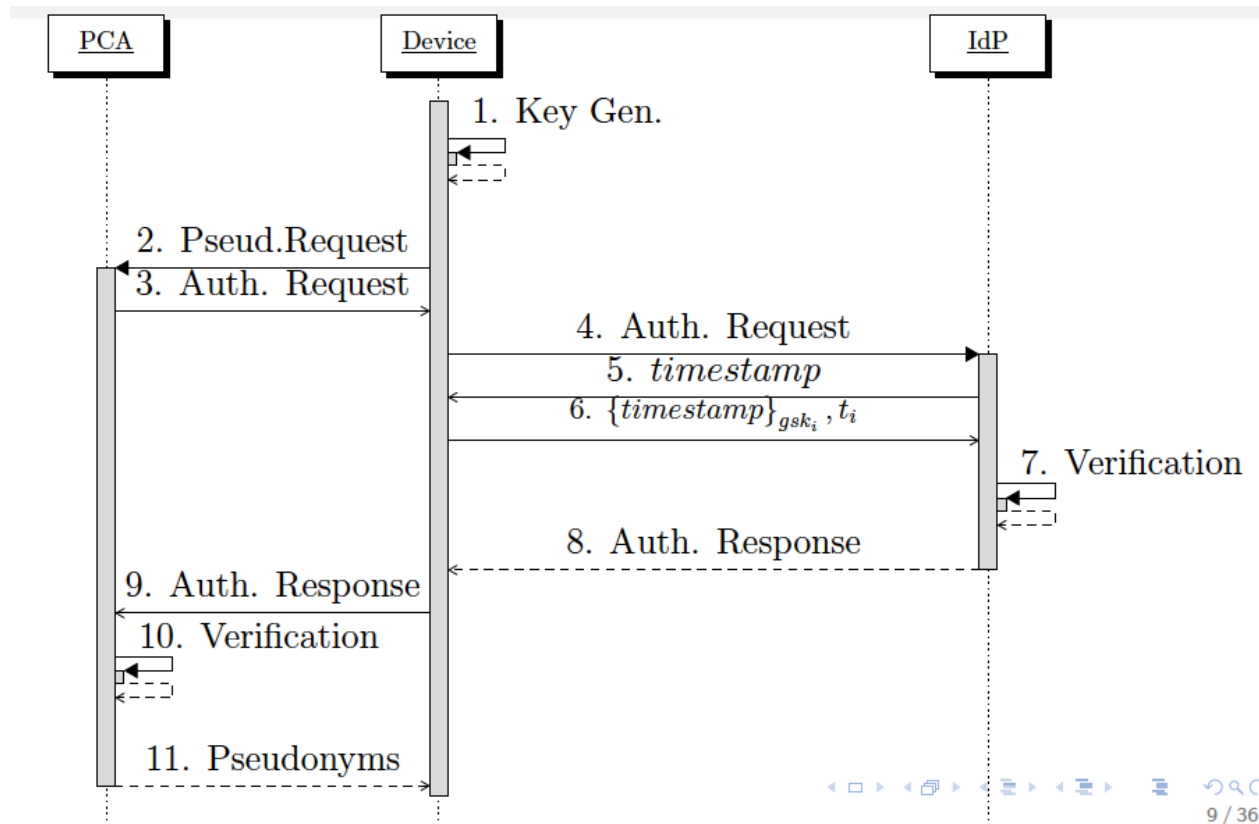




SPPEAR Registration

- Group Manager and group credentials
 - Private key, gsk_i , and group public key, gpk
- The Group Manager generates one Authorization Token Dispenser
 - $D_{auth} = [t_i, t_{i+1}, \dots, t_N]$
 - $t_i = \{t_{id}, h(user_{id} || task_i || n), task_i, submit/query\}$
 - To participate in a specific task, the device needs to obtain the corresponding token without revealing which task
 - Oblivious Transfer Protocol (Oblivious Transfer with Adaptive Queries, M. Naor and B. Pinkas)

SPPEAR Authentication



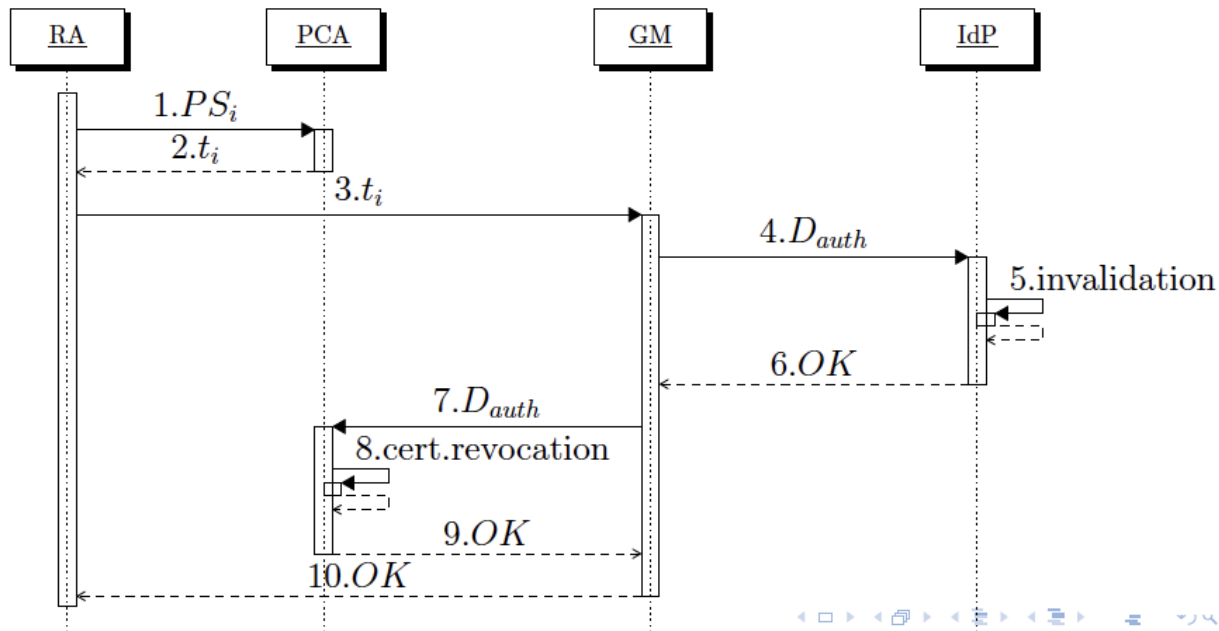


SPPEAR Pseudonym Acquisition

- Pseudonym lifetimes do not overlap
- Pseudonyms used to submit data to SAS
- $s_i = \{v || t || loc || \sigma_{prvKey} || C_i\}$
- $r_i = \{receipt_{id} || task_{id} || time || \sigma_{SAS}\}$
- Or to authenticate queries to SAS
- Note: communications are anonymized with TOR

SPPEAR Device Revocation

- Initiated by a Resolution Authority (RA)
- Anonymity revocation or outright eviction





SPPEAR Analysis

- Confidentiality, integrity (TLS and digital signatures)
- Access control, authorization (GM = PDP and IdP = PEP)
- Sybil-proof (non-overlapping pseudonyms)
- GM does not know the user task(s) (OT for token retrieval)
- Unlinkable and unobservable interactions (TOR)
- Accountability, exculpability (Revocation protocol + interactive mode for BBS)

Honest-but-curious SPPEAR Entities

Honest-but-curious (colluding) entities	Information Leaked	Privacy Implications
GM	-	No sensitive information can be inferred.
IdP	t	The IdP can simply infer that an anonymous user wishes to participate in a task.
PCA	PS, t	The PCA will infer that an anonymous user wishes to receive pseudonyms for a given task.
SAS	s, PS, r	The SAS knows that a given report was submitted for a specific sensing task.
GM, IdP	t, id	The GM and the IdP can infer that a user with a known identity wishes to participate in a specific task.
GM, PCA	t, id, PS	The GM and the PCA can infer that a user with a known identity wishes to participate in a specific task and has received pseudonyms.
GM, SAS	s, PS, r	When the GM and the SAS collude they can infer that a report was submitted by a pseudonymous user.
IdP, PCA	t, PS	These authorities can infer that an anonymous user received pseudonyms for a specific task.
PCA, SAS	t, PS, s, r	The PCA and the SAS can infer that an anonymous user received pseudonyms for a specific task and has submitted a report.
GM, PCA, SAS	all	Full de-anonymization of the user, the task she participates in and the reports she has submitted.

Table: Honest-but-curious entities

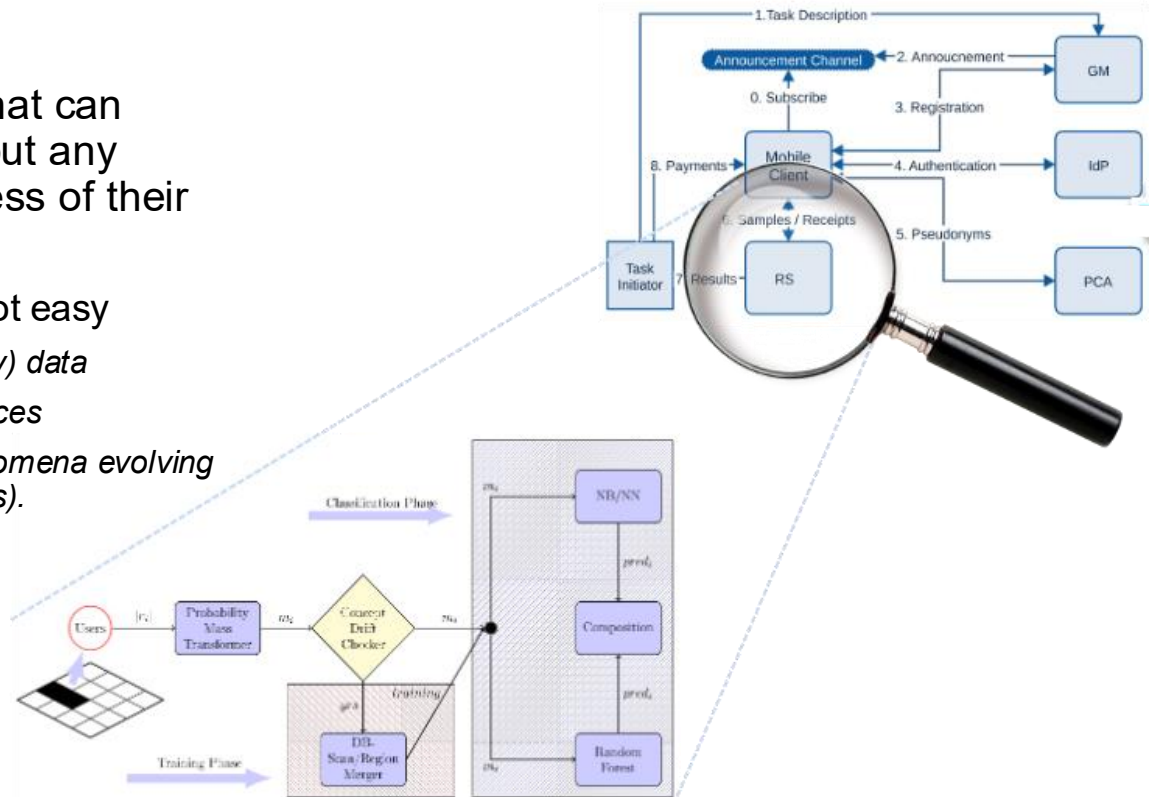
How to ensure data quality in PS?

- Quality of Information
 - Truth Discovery
 - > *Majority voting and averaging not always effective*
 - Do not consider colluding participants and differences in trustworthiness/reliability
 - Trust Framework
 - > *Trust computed based on the participant's past trust and his/her current report*
 - > *Qol usually computed by aggregating the trust score of the participants reporting it*
 - > *Data centric trust*
 - **M. Raya, P. Papadimitratos, V. Gligor, and J. Hubaux, "On Data-Centric Trust Establishment in Ephemeral Ad hoc Networks," *IEEE Conference on Computer Communications (IEEE INFOCOM)*, Phoenix, AZ, USA, pp. 1238-1246, April 2008**

F. Restuccia et. al "Quality of information in mobile crowdsensing: Survey and research challenges", 2017.

Privacy-preserving trustworthy PS SHIELD

- Goal: Design of a mechanism that can assess and sift faulty data without any assumption on the trustworthiness of their source.
 - Designing such mechanisms is not easy
 - > Requires the fusion of (contradictory) data
 - > Originating from untrustworthy sources
 - > Describing dynamic/uncertain phenomena evolving spatio-temporally (i.e., concept drifts).
- Thread modes
 - Users: (Colluding) malicious
 - System: Honest-but-curious

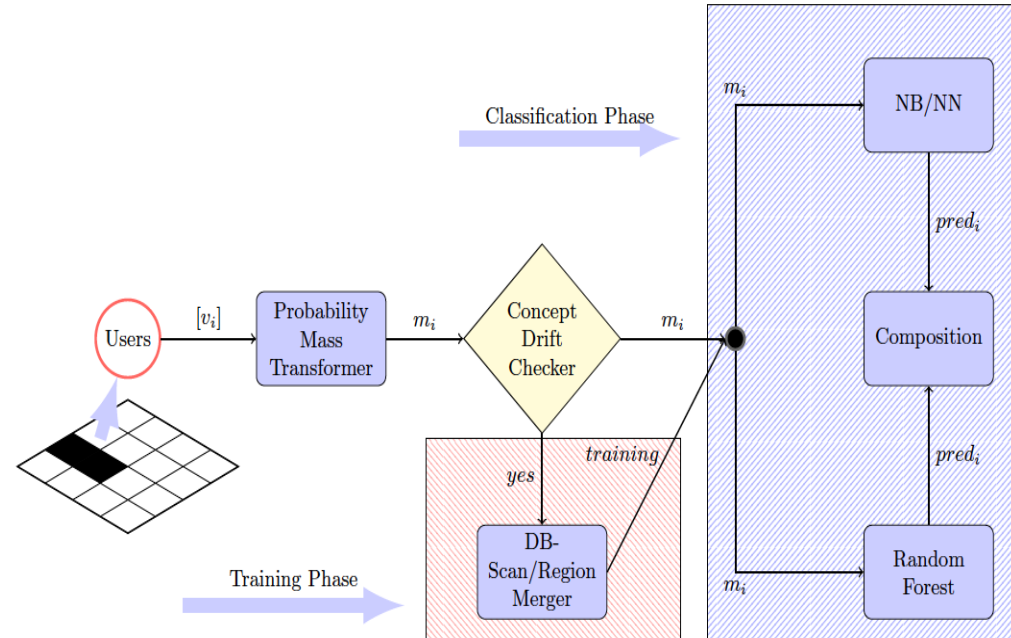


S. Gisdakis, T. Giannetsos, and P. Papadimitratos, "SHIELD: A Data Verification Framework for Participatory Sensing Systems," *ACM Conference on Security & Privacy in Wireless and Mobile Networks (ACM WiSec)*, 2015

Privacy-preserving trustworthy PS (cont'd)

SHIELD

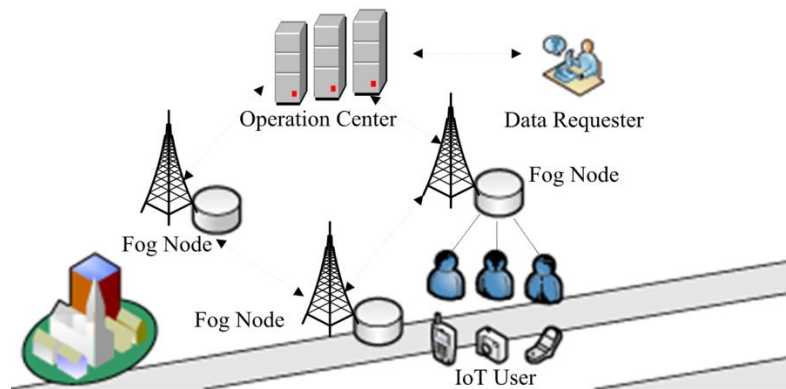
- How to handle the evidence?
 - Dempster-Shafer Theory of Evidence
- Phases
 - Bootstrapping (labelling faulty data based on the structure)
 - > *Using DBSCAN, which relies on honest majorities.*
 - Region Merging and Training
 - > *Region merging is based on the statistical similarity of empirical distributions using a two-sample Kolmogorov-Smirnov (K-S) test.*
 - Classification
 - > *Ensemble of classifiers comprising random forest, naive Bayes and kNN, based on a majority-based decision*



Privacy-preserving Truth Discovery

RPPTD

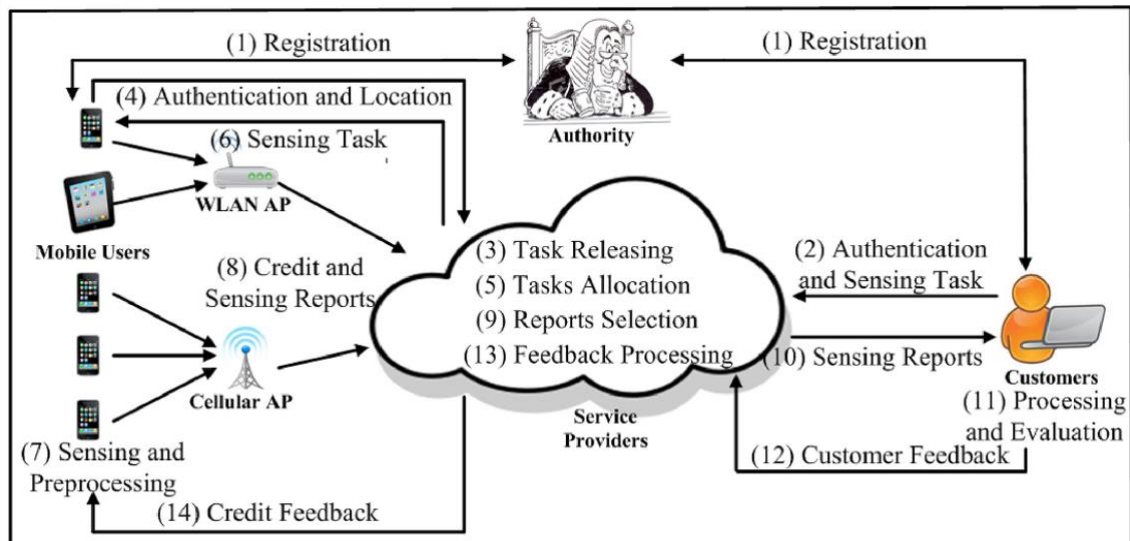
- How to extract the most reliable data while preserving user privacy without the need for a trusted third party and non-colluding platforms
- Threat model
 - Users: follow the protocol, may submit inaccurate data
 - System: Operation center and the fog nodes Honest-but-curious (semi-honest)
- Security and privacy goals
 - No one has access to the user's private data except the user
 - Resistance against modification, impersonation, replay
 - Resistance against human-factor-aware differential aggregation attack



J. Chen, Y. Liu, Y. Xiang and K. Sood, "RPPTD: Robust Privacy-preserving Truth Discovery Scheme," in IEEE Systems Journal, 2021

Privacy-preserving PS with Reputation SPOON

- Privacy-preserving task allocation
 - > *Task initiators and users location privacy from the SP point of view*
 - Privacy-preserving credit management mechanism
- Building Blocks
 - BBS+ signature
 - Matrix multiplication
 - Proxy re-encryption
- Provides greedy user tracking

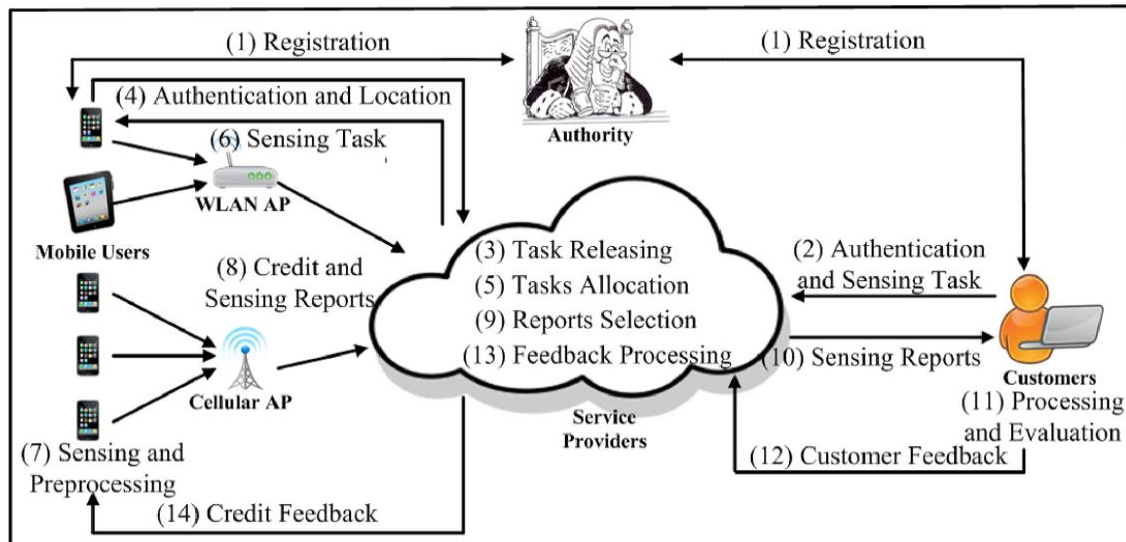


J. Ni, K. Zhang, Q. Xia, X. Lin and X. S. Shen, "Enabling Strong Privacy Preservation and Accurate Task Allocation for Mobile Crowdsensing," in IEEE Transactions on Mobile Computing, June 2020.

Privacy-preserving PS with Reputation

SPOON

- Service Provider
 - Honest but curious + does not collude with the users because of its reputation
- Users
 - Interested in customers
 - Interested in knowing about other users in the same task
 - Greedy to submit more reports but follow the reputation update policy
 - Forge or modify sensing reports
- External Attackers: Eavesdroppers and hackers
- Customers: Fully trusted





Privacy-preserving PS with Reputation SPOON

- Report trustworthiness, computed by the Task Initiators as: $\epsilon_i \in [-\gamma, \gamma]$
 - Partitioning sensing area into grids and computing the weights (w_z) of data in each grid, where $\sum w_z = 1, w_z \in (0, 1]$
 - In each grid (z):
 - > *Computing similarity of a report:* $V_{i,z} \in [-\gamma, \gamma]$
 - γ = The maximum of trust level in a task
 - > *Computing trust level of a report:* $\epsilon_{i,z} = \left(\frac{V_{i,z} Q_i}{\sum_{i \in Z} V_{i,z} Q_i} \right) w_z \gamma$
 - Q_i : User chosen reputation threshold
 - User trust level
 - > *Data in the report belong to the same grid:* $\epsilon_i = \epsilon_{i,z}$
 - > *Data in the report belong to the multiple grid:* $\epsilon_i = AVE(\epsilon_{i,z})$
- User credit, computed by the Service Provider: $\theta_i = INT(\epsilon_i Q_i)$
- Reputation update: $P_i + \theta_i$



Autumn 2025

Discussion leads

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